1.2 Final Report

Thoa Le (a1847933)

Abstract

Introduction and Literature Review

Methods

Data from 1960 to 2024 on Total Live Births (TLB) and Total Fertility Rates (TFR) was pulled from Singapore Department of Statistics (data.gov.sg). The data was then transformed into a tsibble object with three columns: Year, Total Fertility Rates (TFR) and Total Live Births.

The TFR was initially plotted to examine the presence of trends, seasonality, and other notable patterns (Figure 1). The subsequent analysis focuses on the period from 1960 to 2012, a second plot was produced for visualisation and modelling purposes (Figure 2).

There is a long-term decreasing trend over time hence the data was differenced. Viable ARIMA models were found but the KPSS unit root test suggested the data needed to be differenced again where more viable ARIMA models were found. The initial plot decreasing trend suggests a log transformation would be appropriate where more models were found after differencing and twice differencing.

Figure 1. Initial Plot of Total Fertility Rate (TFR)

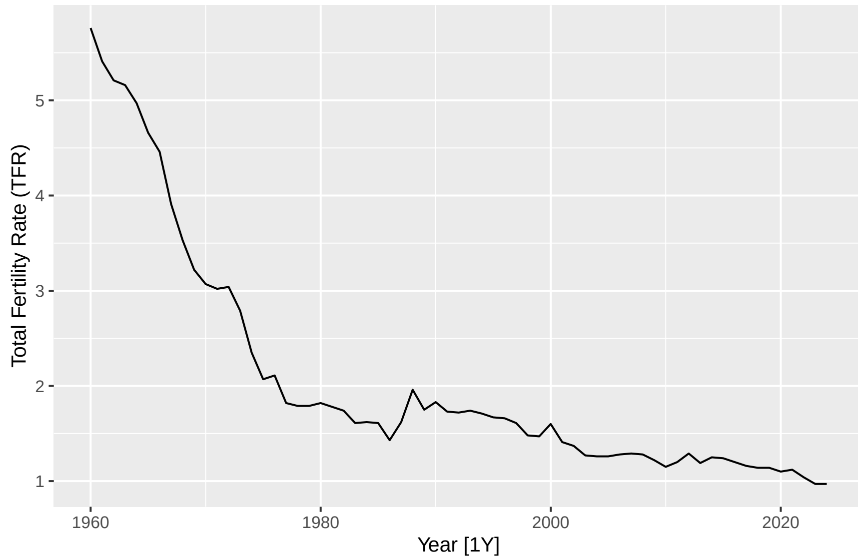


Figure 2. Initial Plot of Total Fertility Rate (TFR) until 2012

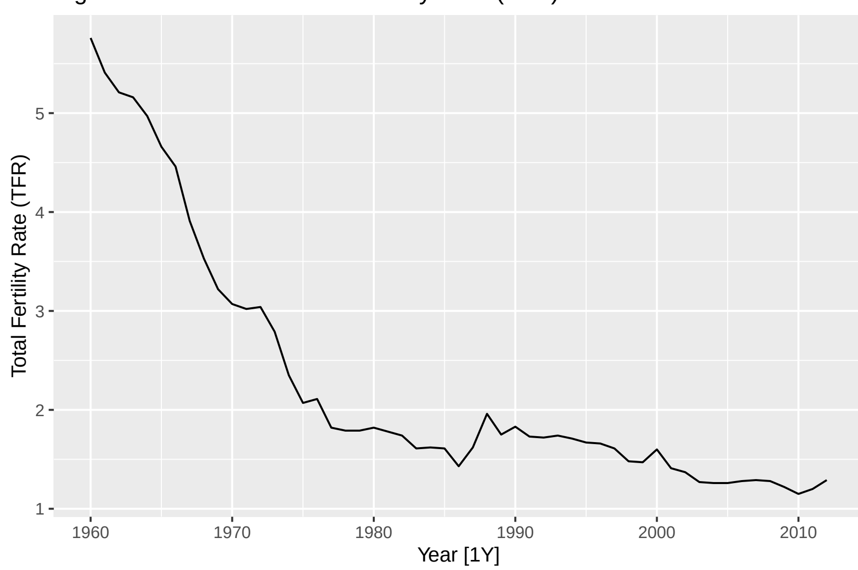
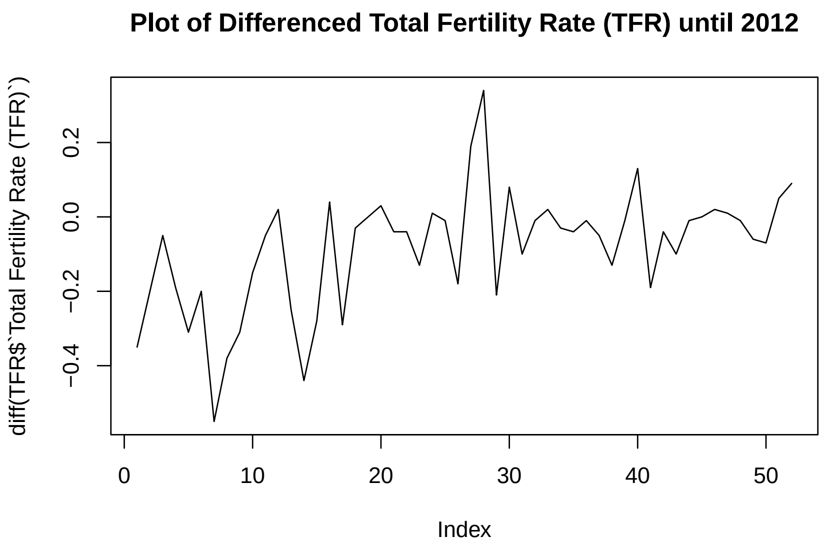
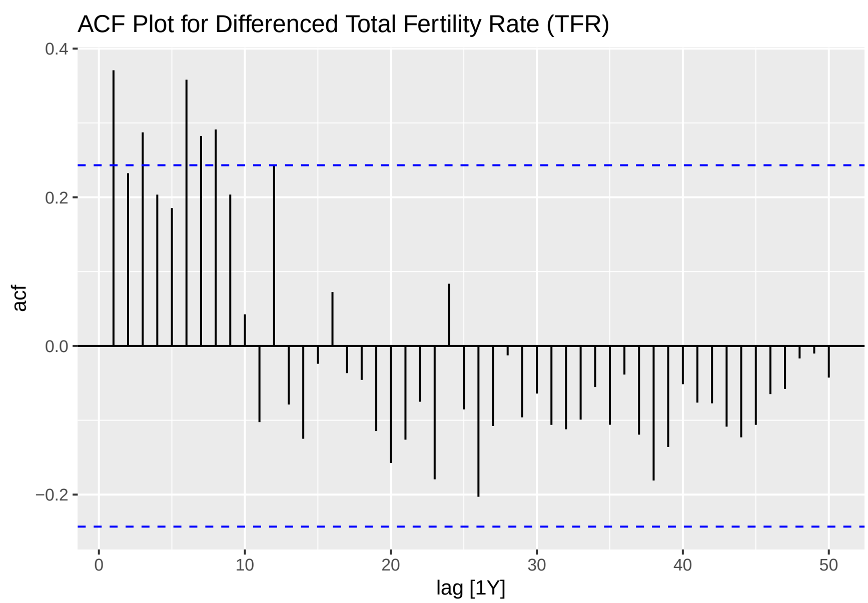
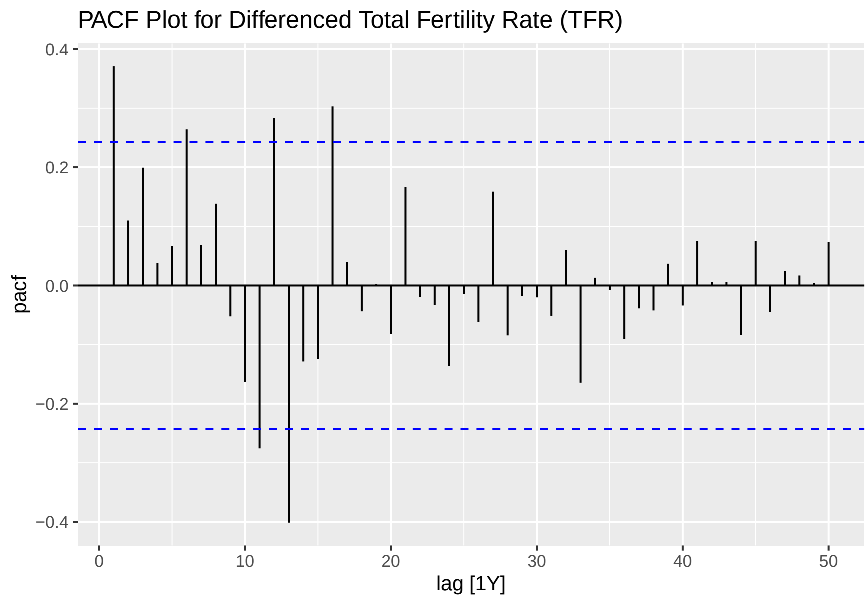


Figure 1 show a long-term decreasing trend over time

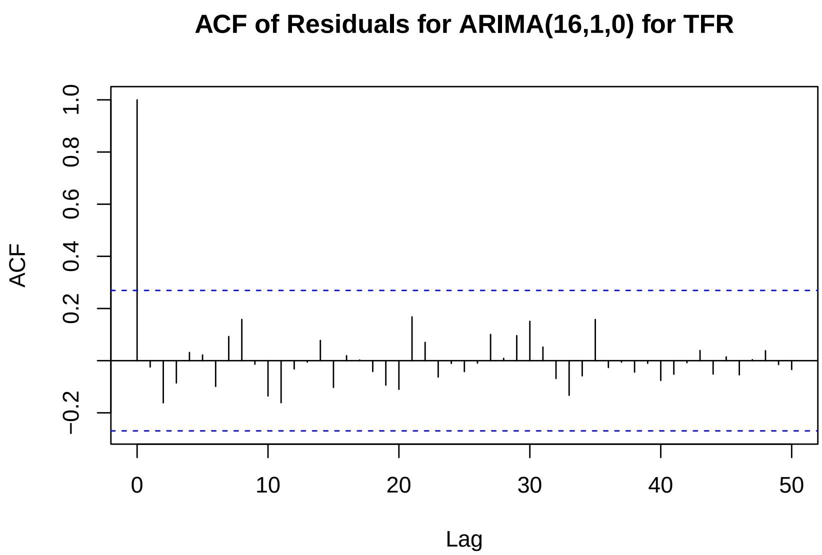


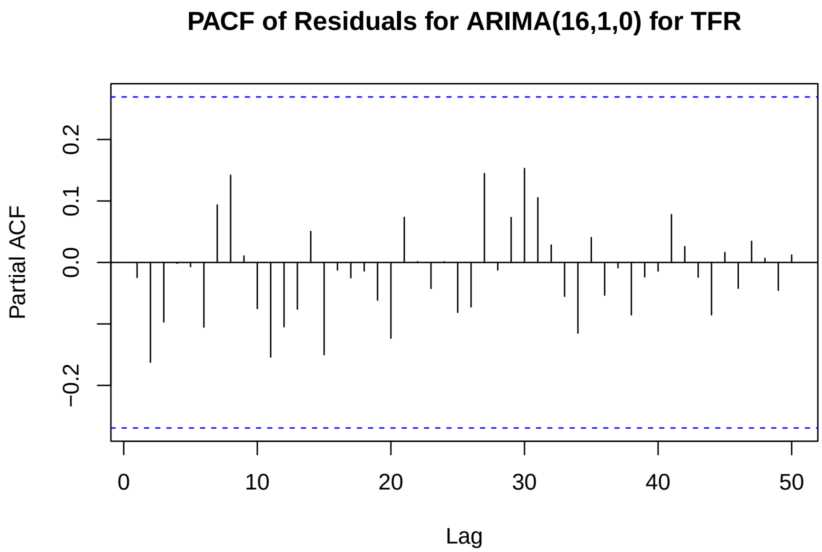

```
##   kpss_stat kpss_pvalue
##   0.8412374   0.0100000
```





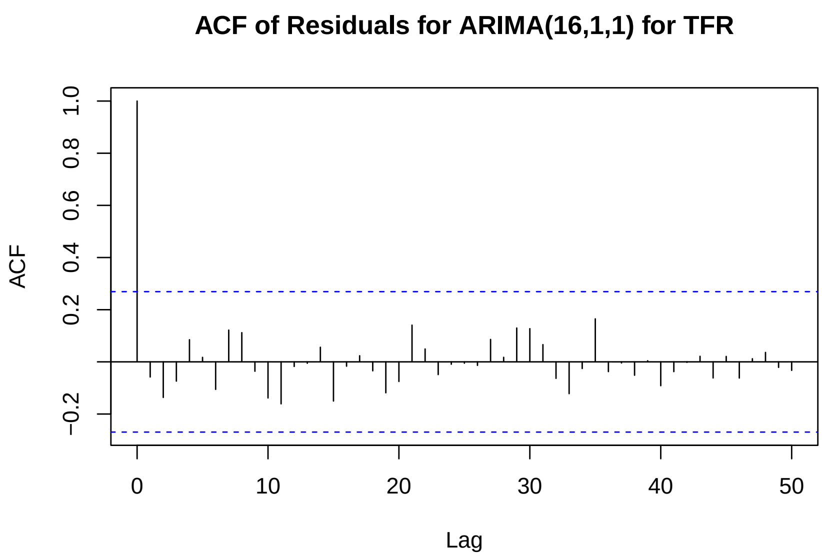

```
## Series: TFR_ts
## ARIMA(16,1,0)
##
## Coefficients:
##          ar1      ar2     ar3      ar4     ar5     ar6     ar7     ar8      ar9
##       0.3647  -0.1061  0.4577  -0.2586  0.2984  0.2452  0.2006  0.1578  -0.0293
## s.e.  0.1240   0.1289  0.1246   0.1128  0.0935  0.0832  0.0952  0.0928   0.0996
##          ar10     ar11    ar12     ar13     ar14     ar15    ar16
##       -0.0306  -0.3594  0.3889  -0.6048  -0.0419  -0.3361  0.5135
## s.e.   0.0973   0.0913  0.1039   0.1066   0.1350   0.1295  0.1397
##
## sigma^2 = 0.009716:  log likelihood = 48.99
## AIC=-63.99   AICc=-45.99   BIC=-30.82
```

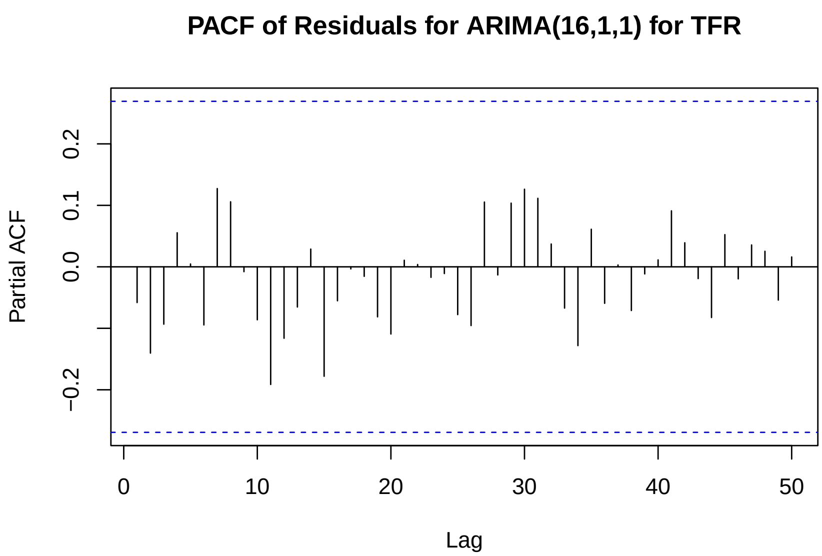




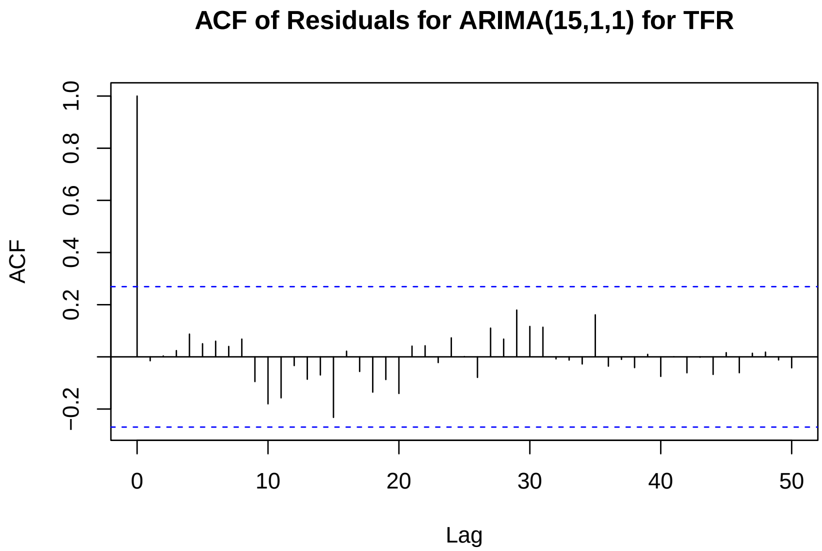
By using differencing, the plots look like the trend has been removed.

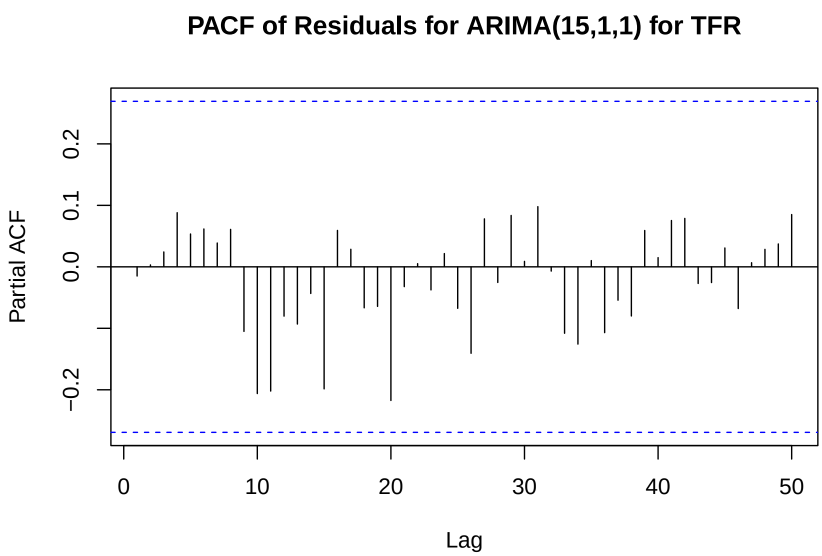
```
## Series: TFR_ts
## ARIMA(16,1,1)
##
## Coefficients:
##           ar1      ar2     ar3      ar4     ar5     ar6     ar7     ar8     ar9
##        0.1420  -0.0490  0.4206  -0.2157  0.2908  0.2807  0.2588  0.2201  0.0473
## s.e.   0.3235   0.1373  0.1329   0.1289  0.0959  0.0994  0.1320  0.1392  0.1541
##          ar10     ar11    ar12     ar13     ar14     ar15    ar16     ma1
##       -0.0052  -0.3380  0.3253  -0.5307  -0.1537  -0.3561  0.4645  0.2959
## s.e.   0.1108   0.0988  0.1375   0.1468   0.1938   0.1269  0.1787  0.3855
##
## sigma^2 = 0.009861:  log likelihood = 49.19
## AIC=-62.39   AICc=-41.66   BIC=-27.27
```



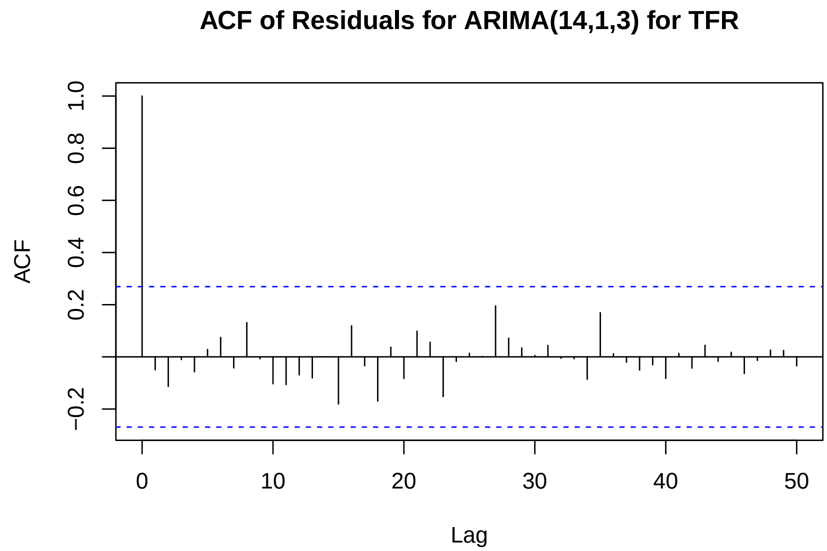


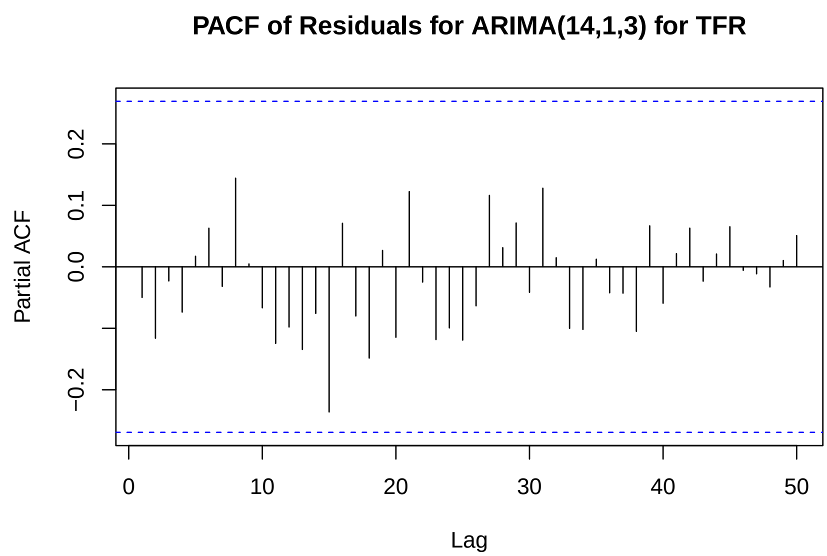
```
## Series: TFR_ts
## ARIMA(15,1,1)
##
## Coefficients:
##            ar1      ar2     ar3      ar4     ar5     ar6     ar7     ar8     ar9
##        -0.3687  -0.0438  0.1991  -0.0219  0.1685  0.3810  0.4689  0.4961  0.3172
## s.e.    0.1653   0.1388  0.1176   0.1129  0.1060  0.1085  0.1178  0.1170  0.1210
##          ar10      ar11    ar12     ar13     ar14     ar15     ma1
##        0.1567  -0.2436  0.1949  -0.3159  -0.3817  -0.4255  0.7608
## s.e.   0.1154   0.1134  0.1377   0.1301   0.1453   0.1472  0.1461
##
## sigma^2 = 0.01082:  log likelihood = 47.13
## AIC=-60.26   AICc=-42.26   BIC=-27.09
```

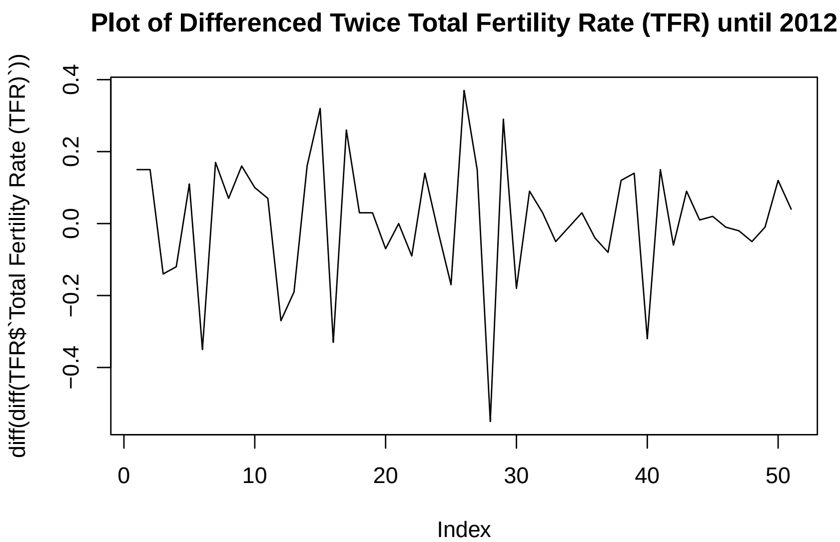




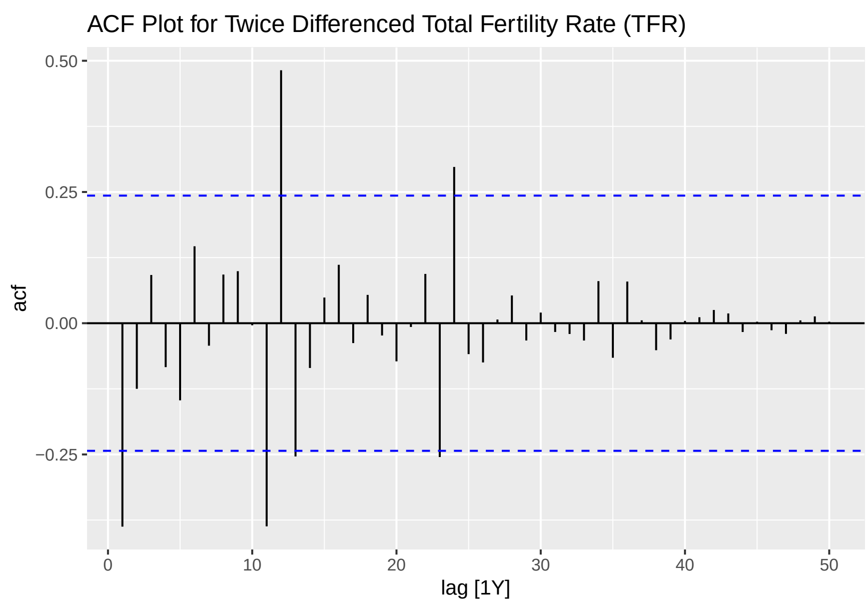
```
## Series: TFR_ts
## ARIMA(14,1,3)
##
## Coefficients:
##          ar1      ar2      ar3     ar4     ar5     ar6     ar7     ar8     ar9
##       0.4303  -0.2164  -0.2058  0.0626  0.1831  0.3127  0.1969  0.2886  0.1127
## s.e.  0.1783   0.1509   0.1244  0.1052  0.1028  0.1023  0.1098  0.1118  0.1131
##         ar10     ar11    ar12     ar13     ar14      ma1     ma2     ma3
##       0.1390  -0.2144  0.4389  -0.5944  -0.1400  -0.1063  0.1280  0.8821
## s.e.  0.1101   0.1120  0.1189   0.1402   0.1747   0.1292  0.1209  0.1426
##
## sigma^2 = 0.009349:  log likelihood = 48.49
## AIC=-60.99   AICc=-40.26   BIC=-25.87
```

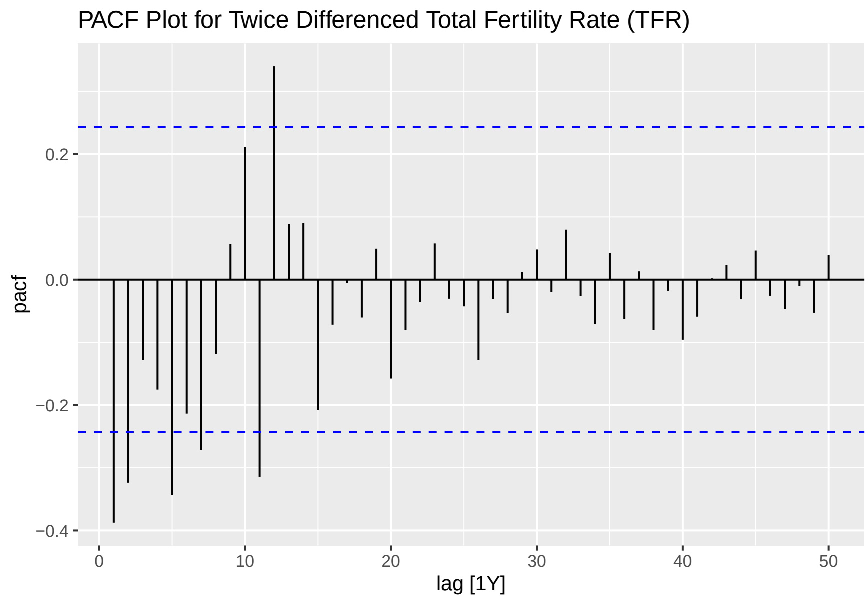




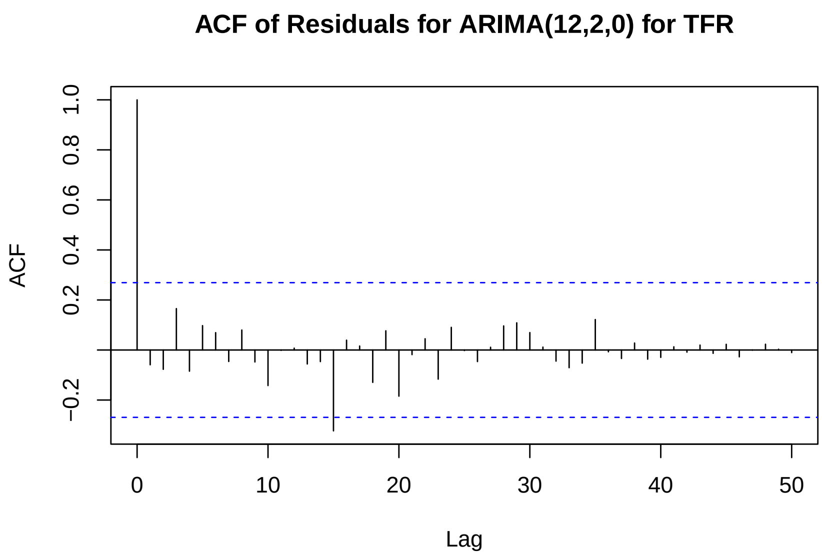


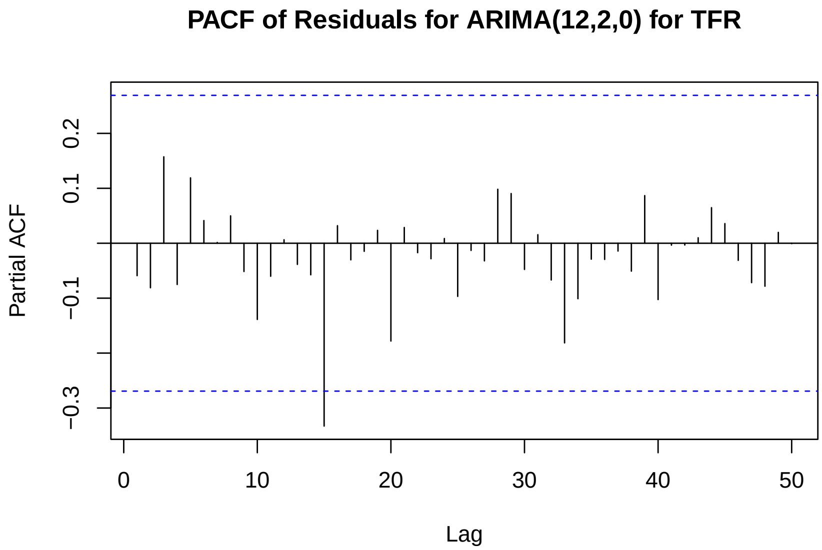

```
##   kpss_stat kpss_pvalue
##  0.04262877  0.10000000
```



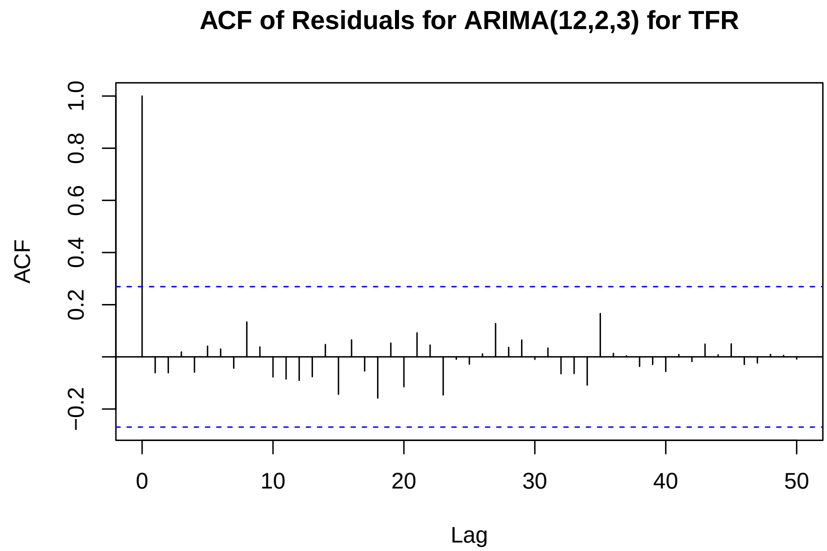


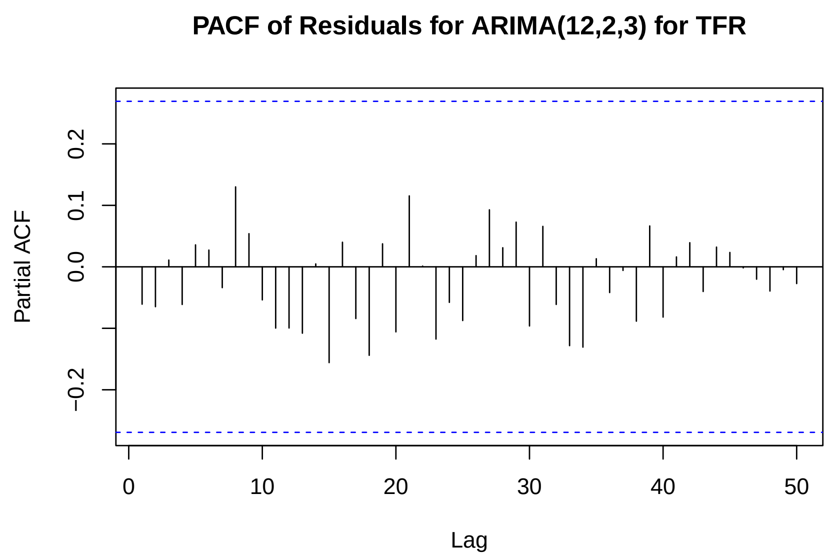

```
## Series: TFR_ts
## ARIMA(12,2,0)
##
## Coefficients:
##           ar1      ar2      ar3      ar4      ar5      ar6     ar7     ar8
##       -0.6234  -0.7441  -0.5813  -0.5700  -0.4163  -0.1741  0.0462  0.3482
## s.e.   0.1203   0.1494   0.1673   0.1676   0.1705   0.1772  0.1846  0.1847
##          ar9    ar10    ar11    ar12
##       0.4206  0.4880  0.1609  0.5040
## s.e.  0.1845  0.1783  0.1540  0.1205
##
## sigma^2 = 0.01286:  log likelihood = 41.78
## AIC=-57.55   AICc=-47.72   BIC=-32.44
```



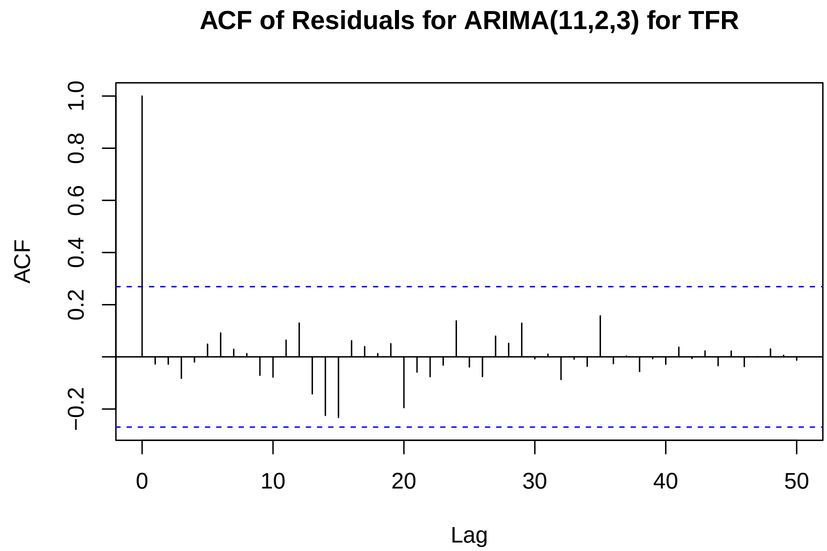

PACF of Residuals for ARIMA(12,2,0) for TFR
Partial ACF
0.2
0.1
−0.1
−0.3
0
10
20
30
40
50
Lag

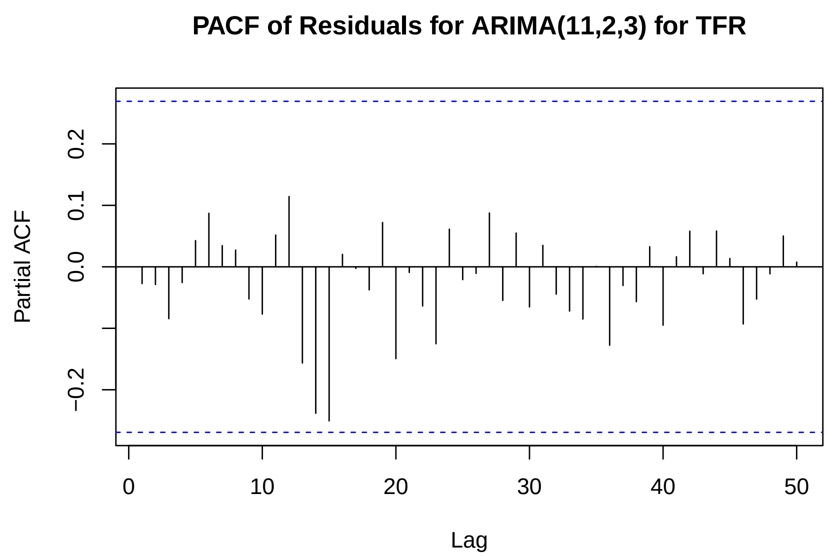
```
## Series: TFR_ts
## ARIMA(12,2,3)
##
## Coefficients:
##           ar1      ar2      ar3      ar4      ar5      ar6      ar7     ar8
##       -0.4622  -0.7606  -0.9273  -0.8808  -0.6967  -0.4124  -0.2312  0.0490
## s.e.   0.1091   0.1266   0.1643   0.2032   0.2376   0.2529   0.2548  0.2404
##          ar9    ar10    ar11    ar12      ma1     ma2     ma3
##       0.1578  0.3268  0.1426  0.6463  -0.1402  0.1411  0.8593
## s.e.  0.2067  0.1645  0.1248  0.1005   0.1237  0.1262  0.1309
##
## sigma^2 = 0.009277:  log likelihood = 47.29
## AIC=-62.57   AICc=-46.57   BIC=-31.66
```



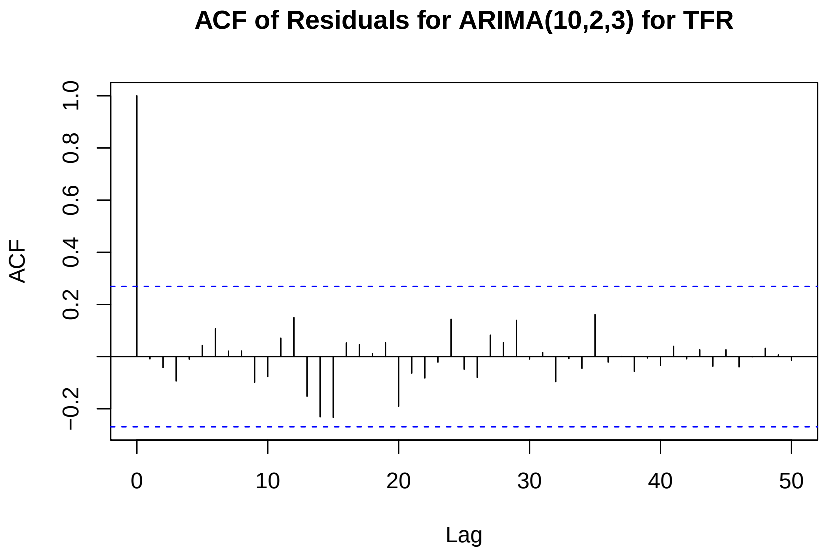


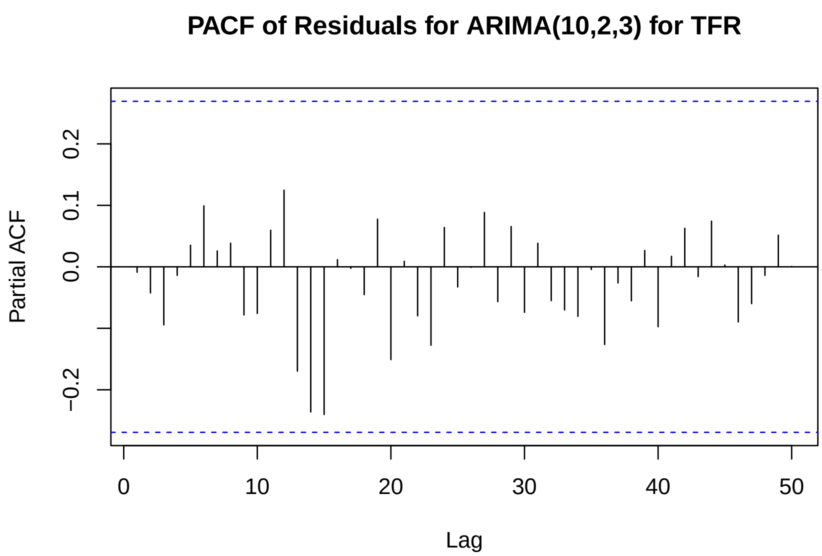
```
## Series: TFR_ts
## ARIMA(11,2,3)
##
## Coefficients:
##           ar1      ar2      ar3      ar4      ar5      ar6     ar7     ar8
##       -1.2619  -0.7197  -0.2948  -0.2528  -0.2564  -0.0704  0.1997  0.4906
## s.e.   0.4667   0.6100   0.5297   0.4227   0.3667   0.3432  0.2914  0.2516
##          ar9    ar10     ar11     ma1      ma2      ma3
##       0.5860  0.4626  -0.0557  0.7978  -0.4841  -0.4123
## s.e.  0.2941  0.3428   0.2956  0.4574   0.3670   0.2321
##
## sigma^2 = 0.01355:  log likelihood = 41.45
## AIC=-52.91   AICc=-39.19   BIC=-23.93
```

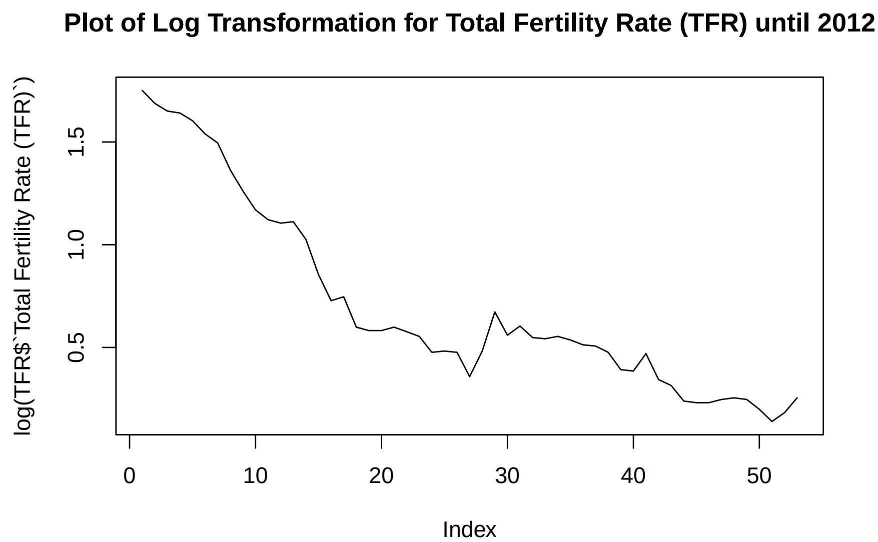


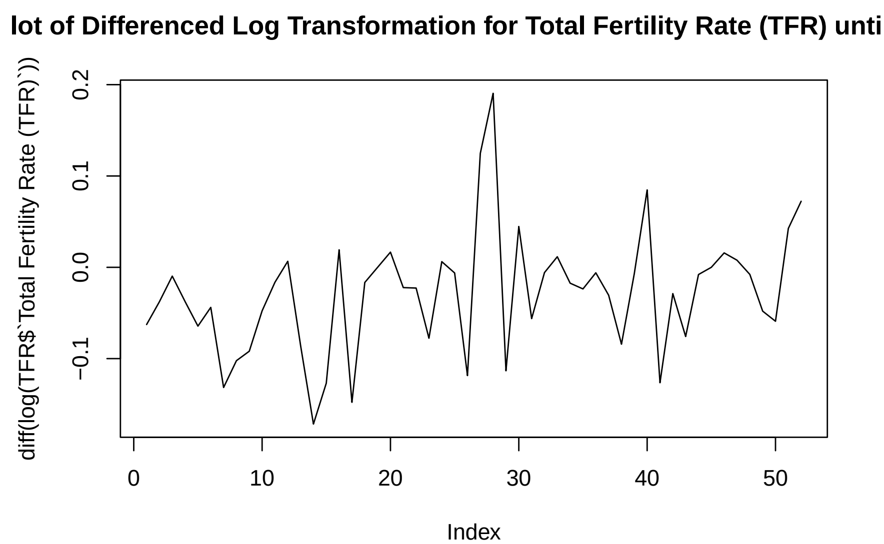


```
## Series: TFR_ts
## ARIMA(10,2,3)
##
## Coefficients:
##            ar1      ar2      ar3      ar4      ar5      ar6     ar7     ar8
##        -1.3103  -0.7996  -0.3813  -0.3141  -0.2988  -0.1007  0.1788  0.4907
## s.e.    0.2181   0.3793   0.4033   0.3580   0.3300   0.3280  0.3015  0.2604
##           ar9    ar10     ma1      ma2      ma3
##        0.6226  0.5298  0.8195  -0.4456  -0.3734
## s.e.   0.2117  0.1231  0.2750   0.3152   0.2122
##
## sigma^2 = 0.01322:  log likelihood = 41.34
## AIC=-54.69   AICc=-43.02   BIC=-27.64
```

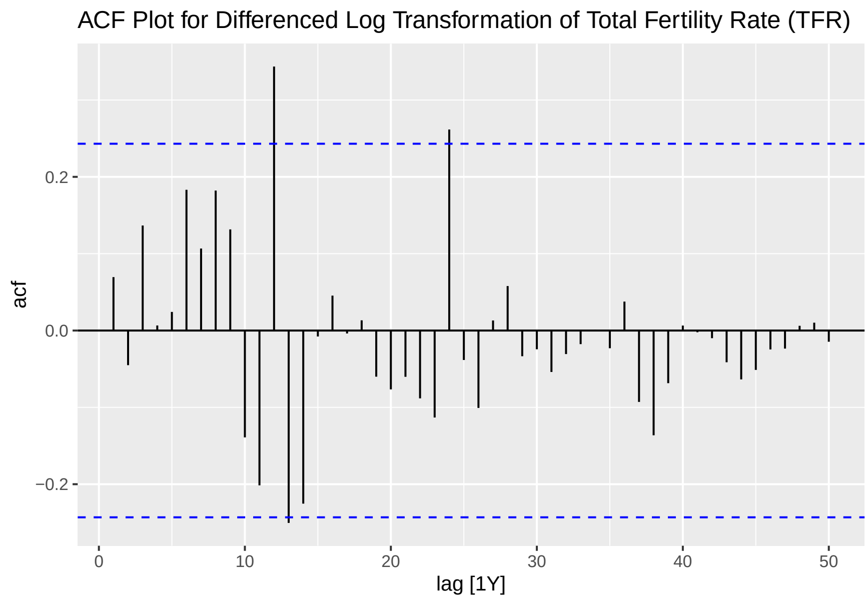


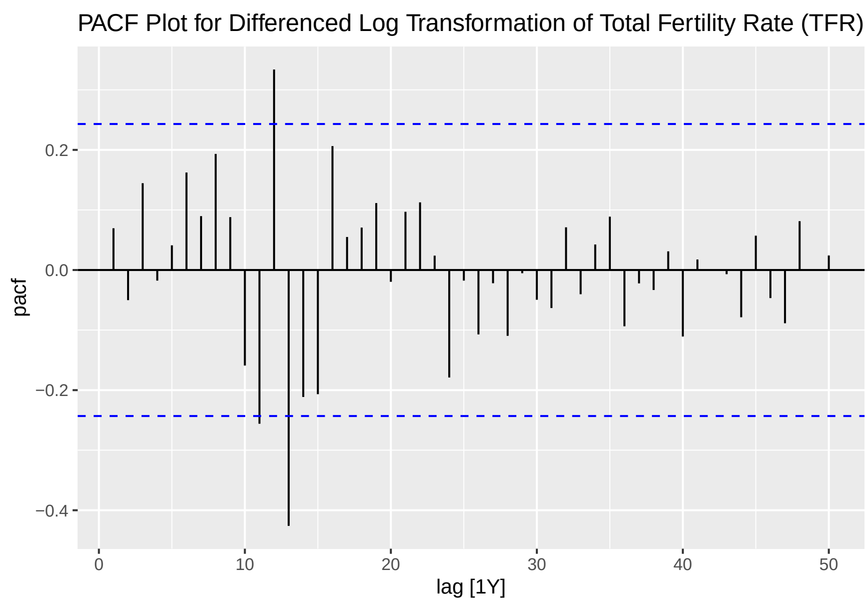






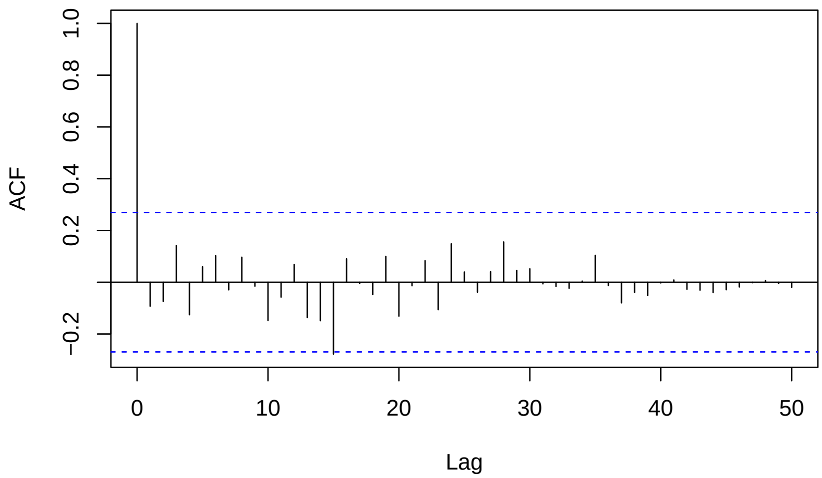

```
##   kpss_stat kpss_pvalue
##  0.50484854  0.04057465
```



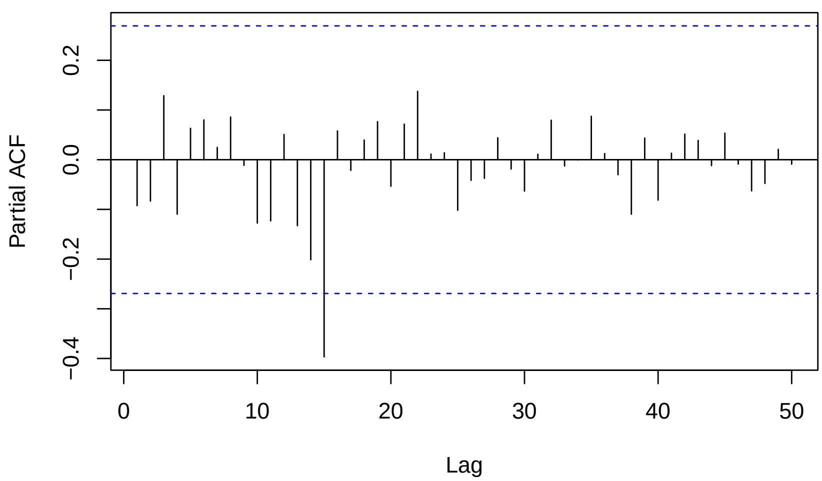



```
## Series: log(TFR_ts)
## ARIMA(13,1,0)
##
## Coefficients:
##          ar1      ar2     ar3     ar4     ar5     ar6     ar7     ar8     ar9
##       0.2629  -0.0799  0.1194  0.0024  0.1349  0.1715  0.1679  0.2392  0.0298
## s.e.  0.1215   0.1145  0.1065  0.1064  0.1079  0.0974  0.1023  0.1036  0.1106
##         ar10     ar11    ar12     ar13
##      -0.0307  -0.2349  0.3587  -0.4375
## s.e.   0.1130   0.1084  0.1170   0.1258
##
## sigma^2 = 0.00323:  log likelihood = 79.9
## AIC=-131.8   AICc=-120.45   BIC=-104.48
```

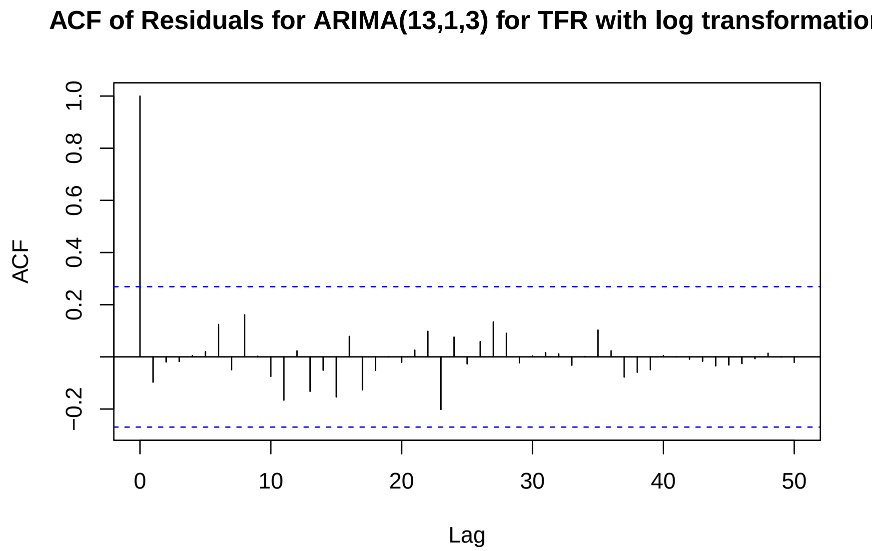
ACF of Residuals for ARIMA(13,1,0) for TFR with log transformation

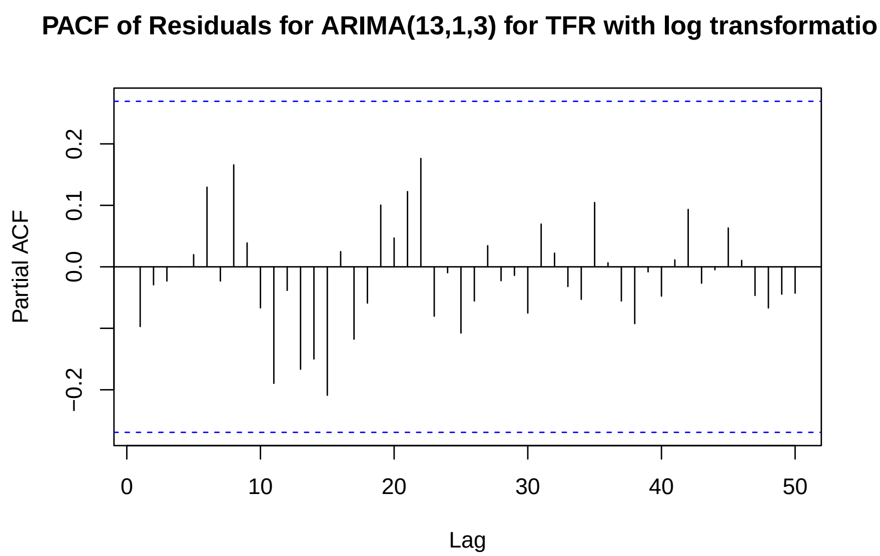


PACF of Residuals for ARIMA(13,1,0) for TFR with log transformatio

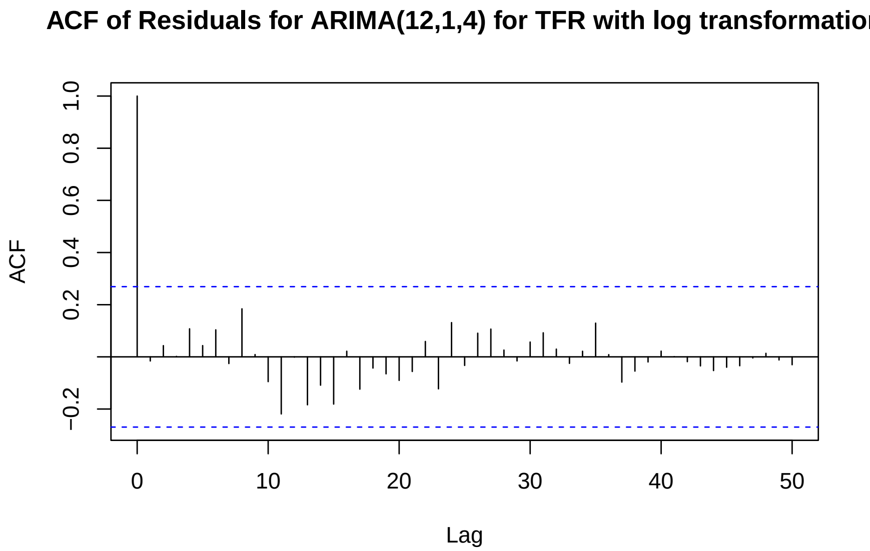


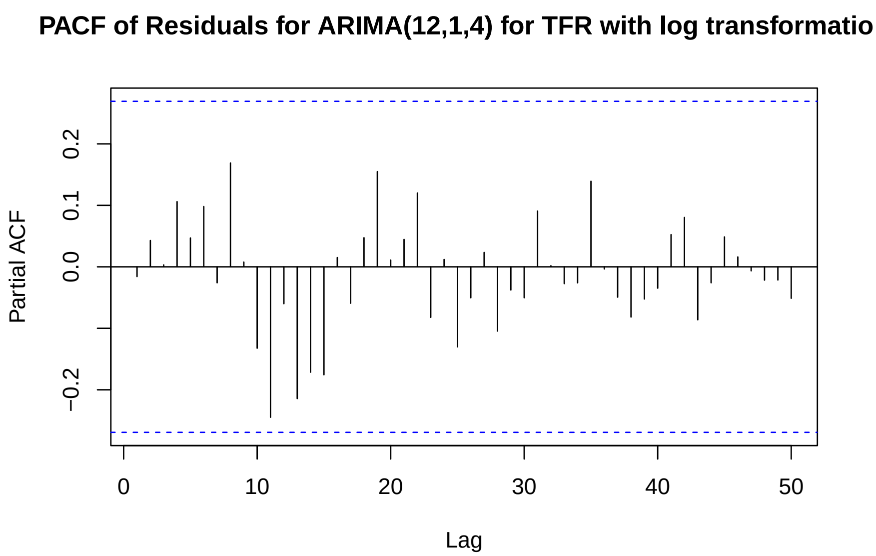
```
## Series: log(TFR_ts)
## ARIMA(13,1,3)
##
## Coefficients:
##          ar1      ar2      ar3      ar4     ar5     ar6     ar7     ar8     ar9
##       0.2783  -0.2434  -0.2961  -0.0013  0.1400  0.2218  0.1567  0.2497  0.0839
## s.e.  0.1256   0.1019   0.1096   0.1094  0.1091  0.0975  0.1024  0.1087  0.1147
##         ar10     ar11    ar12     ar13     ma1     ma2     ma3
##       0.0722  -0.0719  0.4674  -0.5235  0.0903  0.1376  0.9749
## s.e.  0.1170   0.1097  0.1005   0.1256  0.0271  0.1085  0.1089
##
## sigma^2 = 0.00232:  log likelihood = 85
## AIC=-136   AICc=-118   BIC=-102.83
```



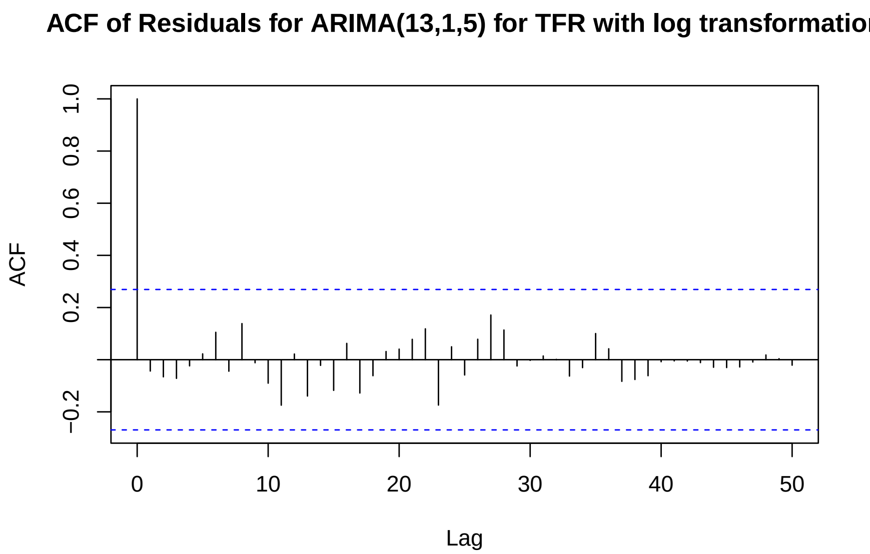


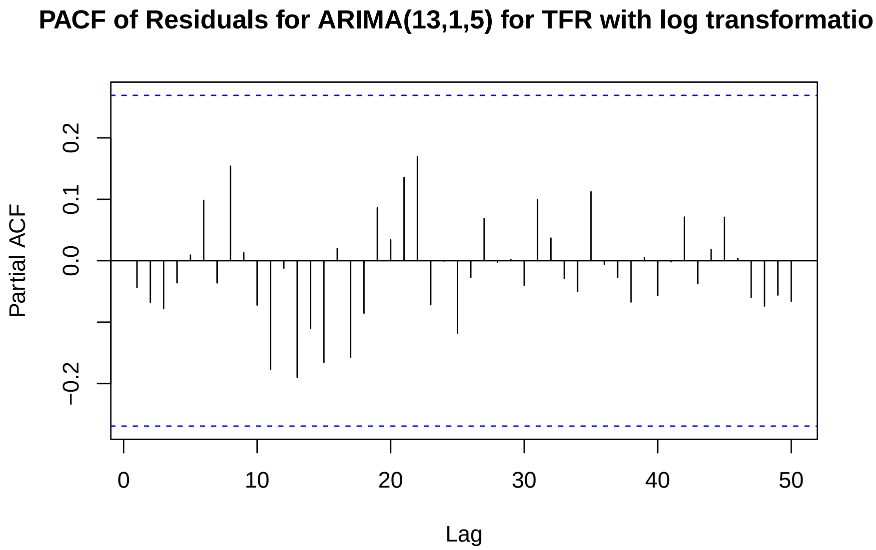
```
## Series: log(TFR_ts)
## ARIMA(12,1,4)
##
## Coefficients:
##          ar1      ar2      ar3      ar4     ar5     ar6     ar7     ar8
##      -0.5373  -0.4532  -0.4935  -0.1889  0.0672  0.2162  0.2570  0.3763
## s.e.  0.2145   0.1628   0.1661   0.1646  0.1507  0.1410  0.1533  0.1605
##         ar9    ar10    ar11    ar12     ma1     ma2     ma3     ma4
##      0.3182  0.2235  0.0423  0.4799  0.9446  0.6719  1.2185  0.5063
## s.e. 0.1762  0.1706  0.1626  0.1536  0.2729  0.1973  0.2749  0.2685
##
## sigma^2 = 0.00256:  log likelihood = 82.91
## AIC=-131.83   AICc=-113.83   BIC=-98.66
```



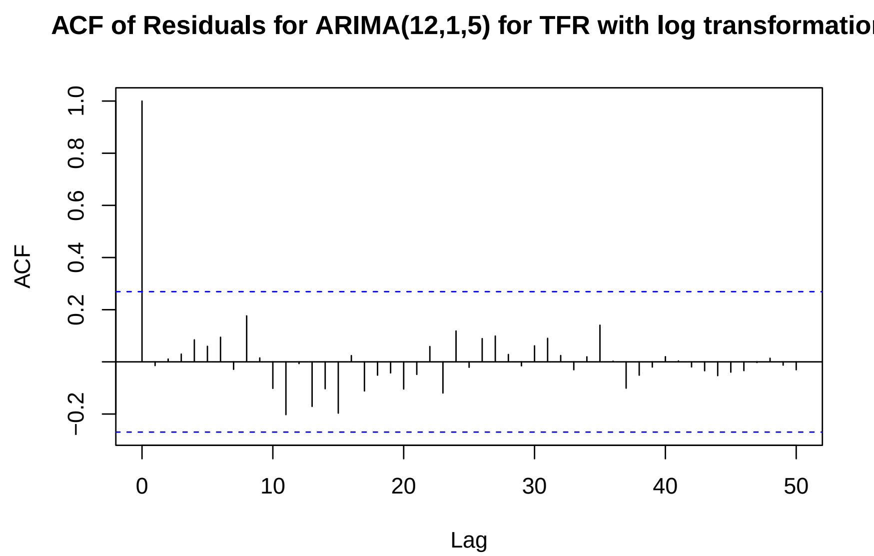


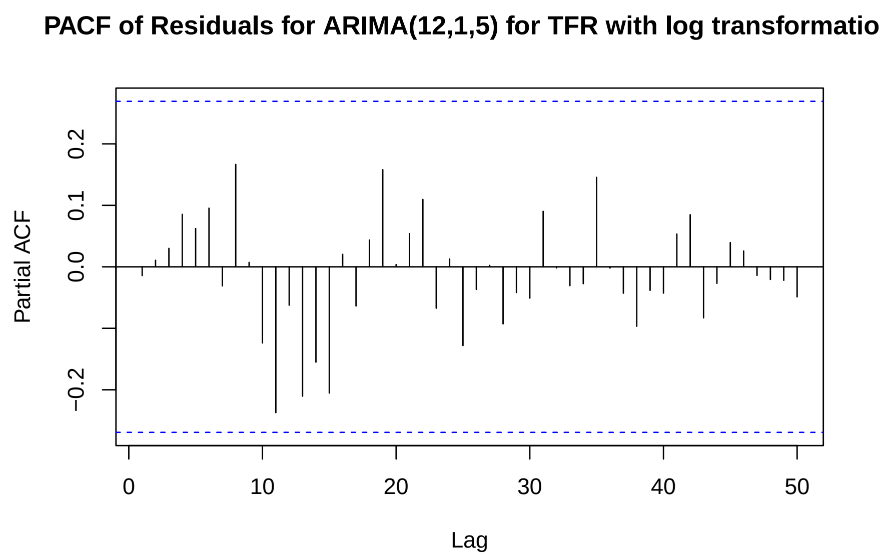
```
## Series: log(TFR_ts)
## ARIMA(13,1,5)
##
## Coefficients:
##          ar1      ar2      ar3     ar4     ar5     ar6     ar7     ar8     ar9
##       0.5505  -0.2059  -0.2132  0.1416  0.1756  0.2193  0.1039  0.2147  0.0036
## s.e.  0.1594   0.1427   0.1180  0.1286  0.1275  0.1045  0.1122  0.1175  0.1168
##         ar10     ar11    ar12     ar13      ma1     ma2     ma3      ma4
##       0.0132  -0.1488  0.4309  -0.6577  -0.2919  0.0354  0.9181  -0.4257
## s.e.  0.1165   0.1123  0.1060   0.1026   0.2578  0.2491  0.1744   0.2493
##           ma5
##       -0.0589
## s.e.   0.2006
##
## sigma^2 = 0.002295:  log likelihood = 85.98
## AIC=-133.96   AICc=-110.21   BIC=-96.89
```

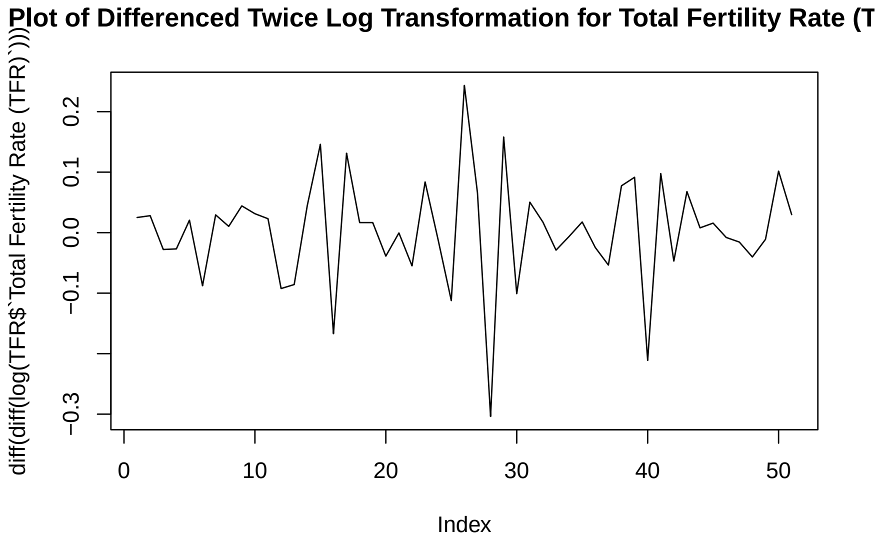



PACF of Residuals for ARIMA(13,1,5) for TFR with log transformatio
Partial ACF
−0.2
0.0
0.1
0.2
0
10
20
30
40
50
Lag

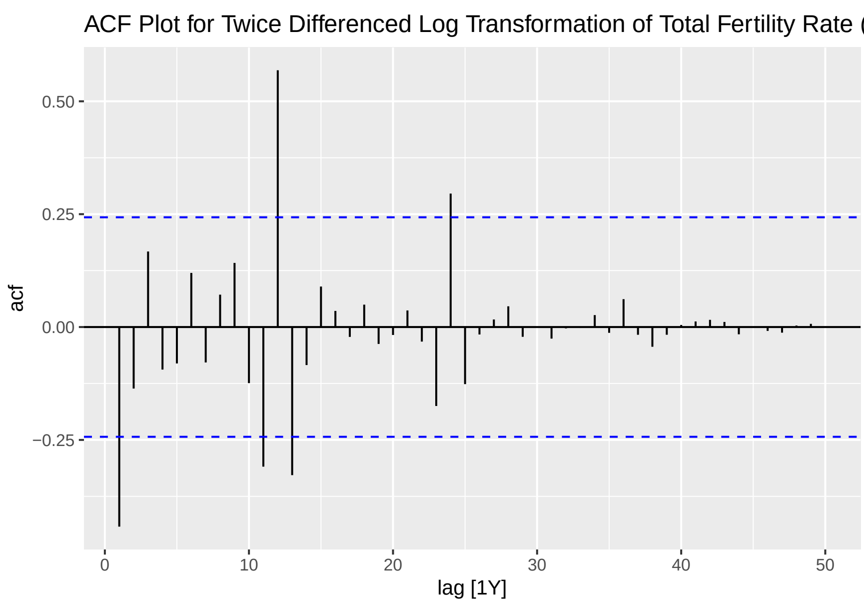
```
## Series: log(TFR_ts)
## ARIMA(12,1,5)
##
## Coefficients:
##           ar1      ar2      ar3      ar4     ar5     ar6     ar7     ar8
##       -0.5680  -0.5453  -0.5384  -0.2383  0.0292  0.2293  0.2780  0.4138
## s.e.   0.2408   0.2656   0.1978   0.1984  0.1779  0.1513  0.1689  0.1873
##          ar9    ar10    ar11    ar12     ma1     ma2     ma3     ma4     ma5
##       0.3539  0.2664  0.0614  0.4901  0.9573  0.7996  1.2817  0.5889  0.1346
## s.e.  0.1990  0.1974  0.1691  0.1594  0.2926  0.3556  0.3255  0.3395  0.2971
##
## sigma^2 = 0.002647:  log likelihood = 83.01
## AIC=-130.02   AICc=-109.3   BIC=-94.9
```

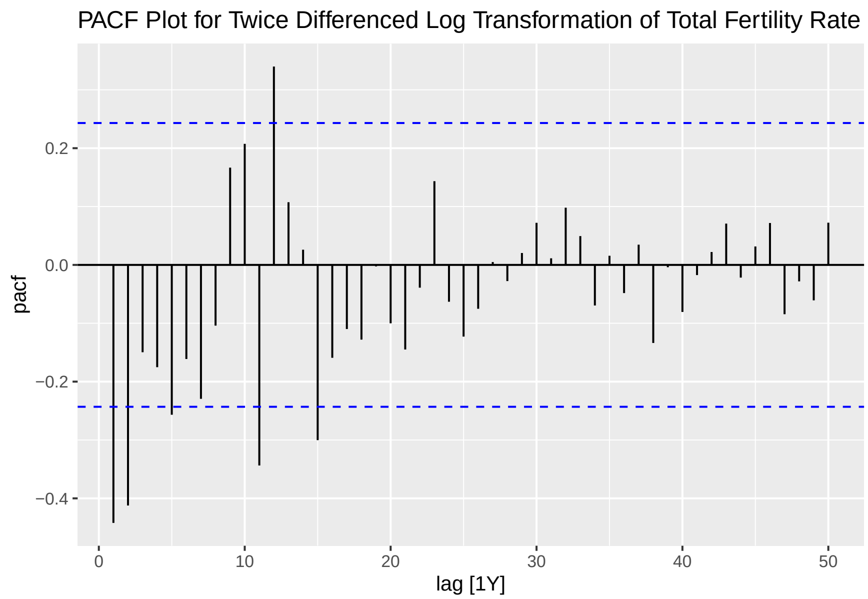




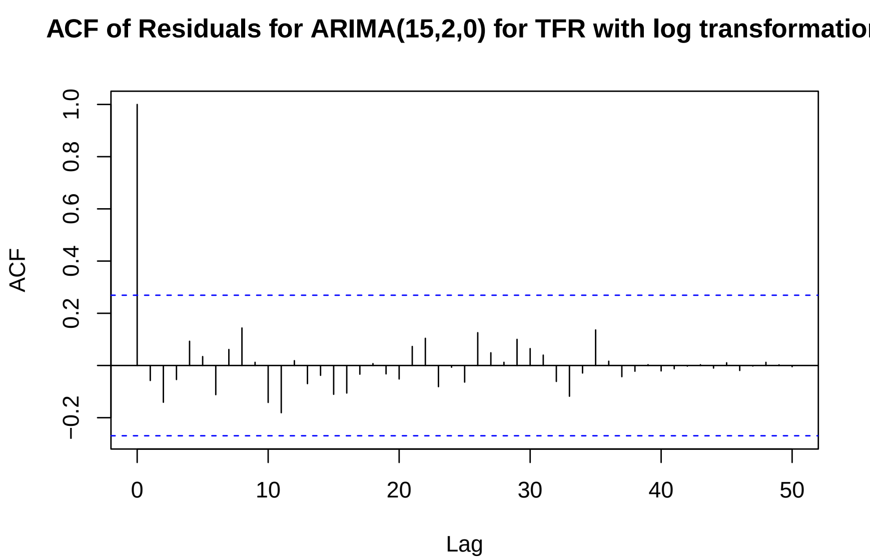


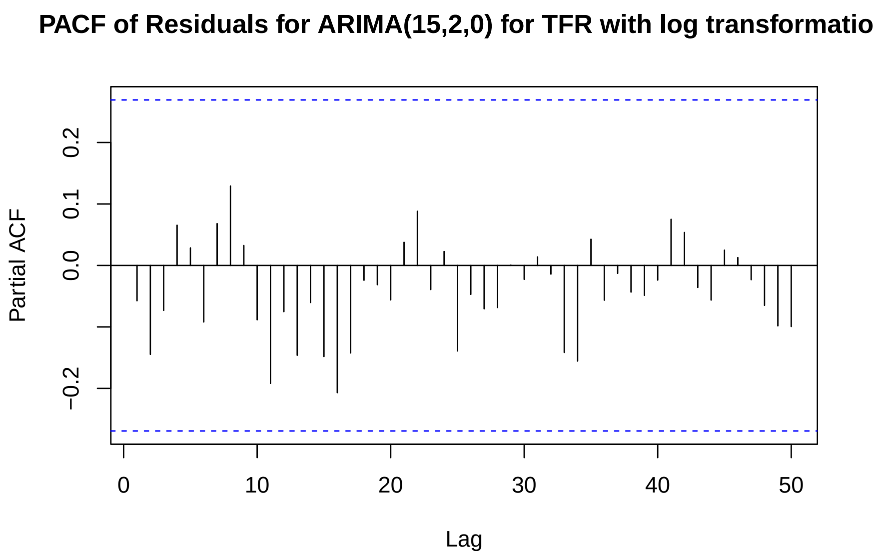

```
##   kpss_stat kpss_pvalue
##  0.04760604  0.10000000
```



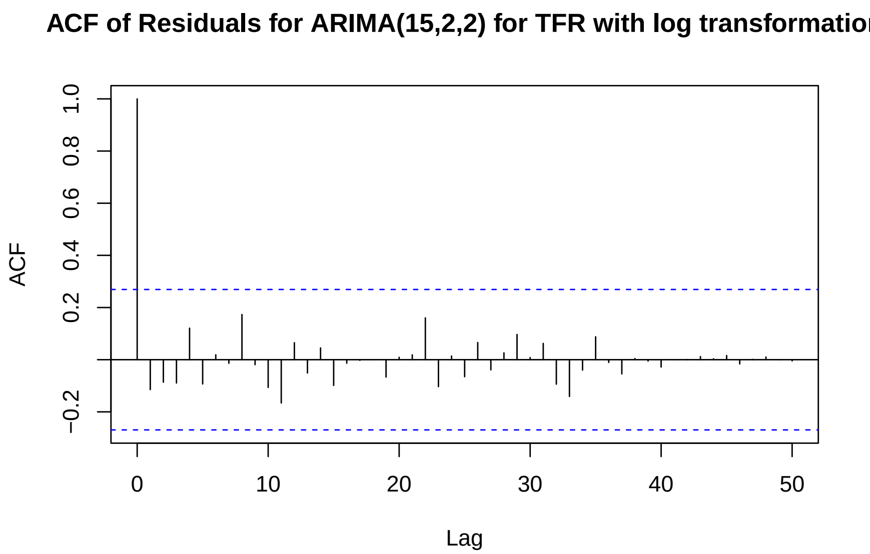


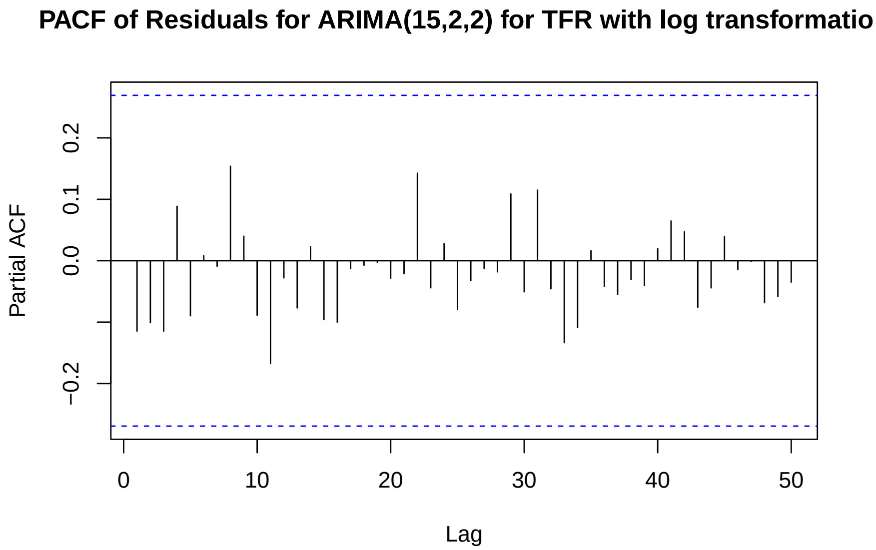

```
## Series: log(TFR_ts)
## ARIMA(15,2,0)
##
## Coefficients:
##            ar1      ar2      ar3      ar4      ar5      ar6     ar7     ar8
##        -0.7137  -0.7004  -0.3372  -0.5826  -0.3331  -0.1103  0.0628  0.2087
## s.e.    0.1218   0.1571   0.1913   0.1824   0.2031   0.2111  0.2146  0.2149
##           ar9    ar10     ar11    ar12     ar13     ar14     ar15
##        0.1864  0.1142  -0.1162  0.3260  -0.1661  -0.2596  -0.4876
## s.e.   0.2152  0.2137   0.2042  0.1855   0.1901   0.1618   0.1289
##
## sigma^2 = 0.002602:  log likelihood = 82.83
## AIC=-133.67   AICc=-117.67   BIC=-102.76
```



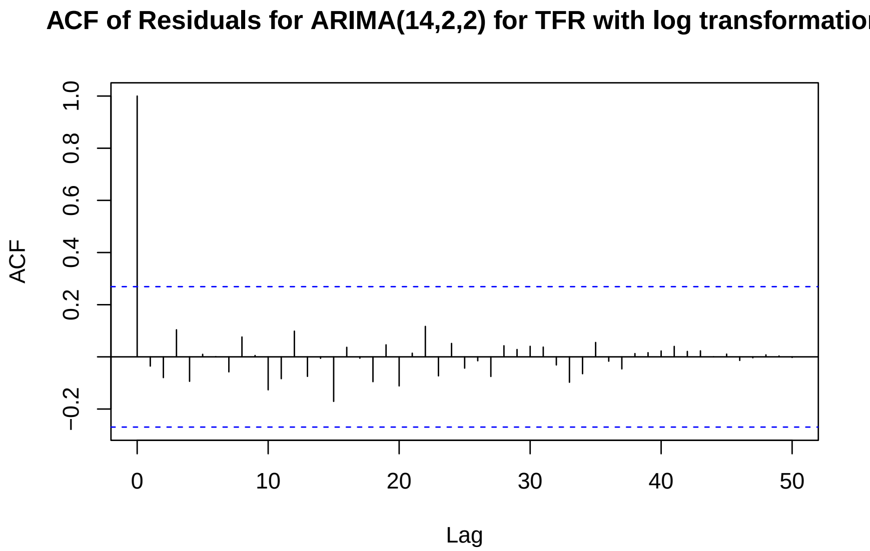

PACF of Residuals for ARIMA(15,2,0) for TFR with log transformatio
Partial ACF
−0.2
0.0
0.1
0.2
0
10
20
30
40
50
Lag

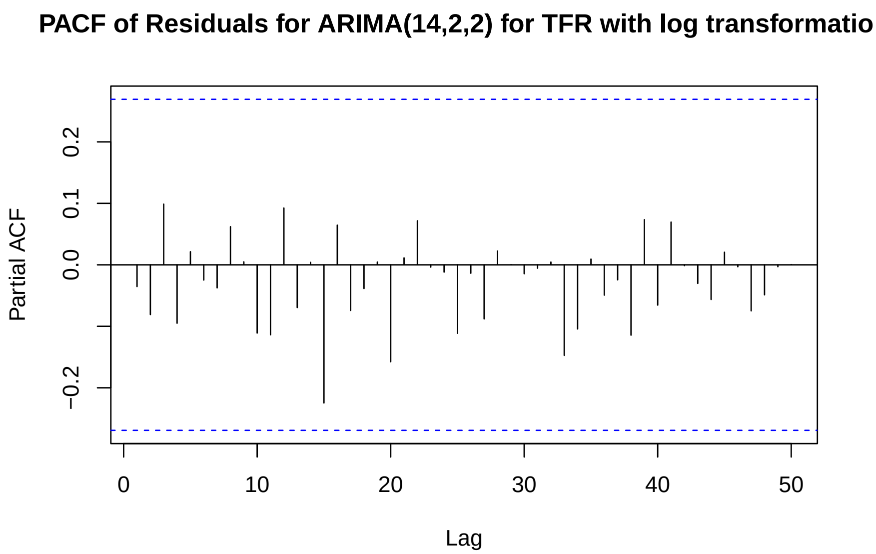
```
## Series: log(TFR_ts)
## ARIMA(15,2,2)
##
## Coefficients:
##           ar1      ar2     ar3      ar4      ar5     ar6     ar7     ar8
##       -0.7683  -0.2252  0.0303  -0.2357  -0.0233  0.2078  0.2915  0.3249
## s.e.   0.1280   0.1620  0.1546   0.1416   0.1384  0.1357  0.1375  0.1305
##          ar9     ar10     ar11    ar12     ar13     ar14     ar15     ma1
##       0.2072  -0.0055  -0.2617  0.1596  -0.2450  -0.5901  -0.5949  0.1513
## s.e.  0.1235   0.1238   0.1168  0.1262   0.1308   0.1250   0.1233  0.1157
##           ma2
##       -0.8290
## s.e.   0.1117
##
## sigma^2 = 0.00213:  log likelihood = 85.77
## AIC=-135.54   AICc=-114.17   BIC=-100.77
```



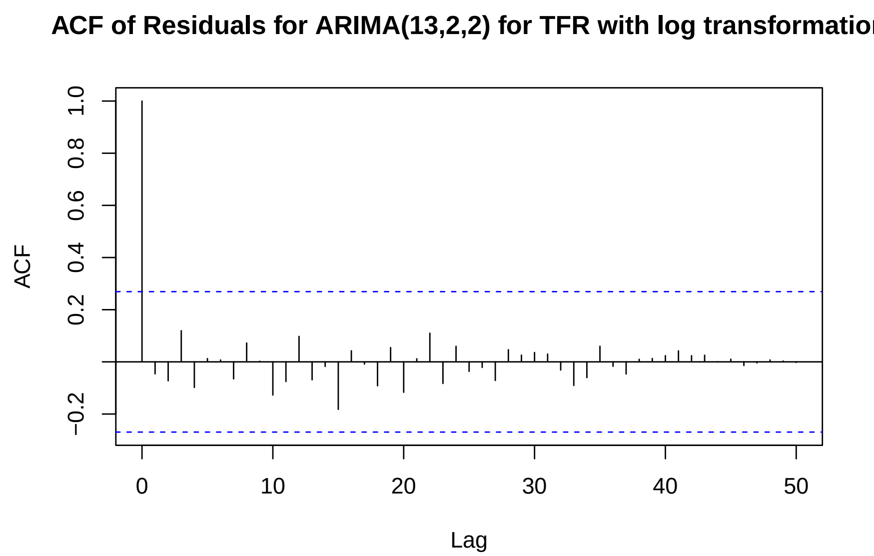

PACF of Residuals for ARIMA(15,2,2) for TFR with log transformatio
Partial ACF
−0.2
0.0
0.1
0.2
0
10
20
30
40
50
Lag

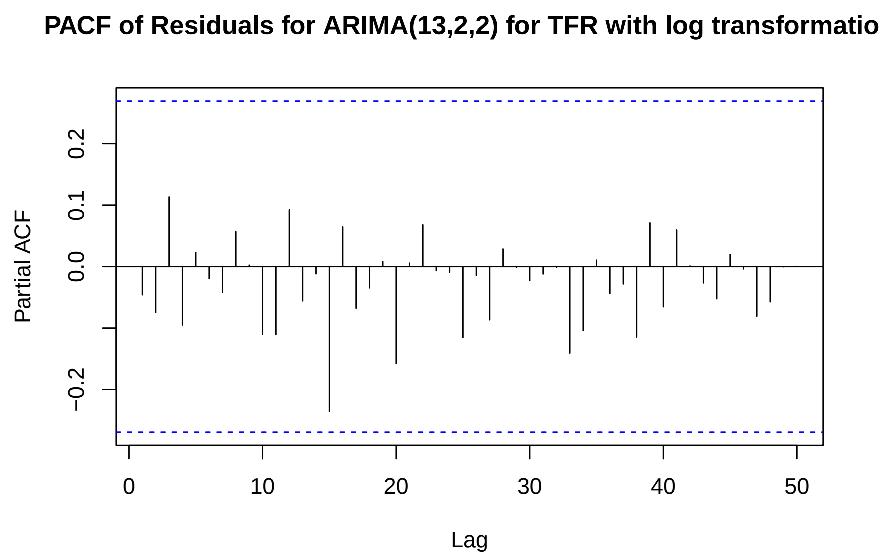
```
## Series: log(TFR_ts)
## ARIMA(14,2,2)
##
## Coefficients:
##          ar1      ar2      ar3      ar4     ar5     ar6     ar7     ar8
##       0.5830  -0.2081  -0.0236  -0.1606  0.0562  0.0850  0.0362  0.1248
## s.e.  0.2416   0.1564   0.1282   0.1162  0.1148  0.1153  0.1125  0.1141
##          ar9     ar10     ar11    ar12     ar13     ar14      ma1     ma2
##      -0.0983  -0.0703  -0.1918  0.4054  -0.5907  -0.0692  -1.5891  0.7422
## s.e.  0.1199   0.1146   0.1108  0.1248   0.1496   0.2117   0.2046  0.1968
##
## sigma^2 = 0.002788:  log likelihood = 82.24
## AIC=-130.49   AICc=-111.94   BIC=-97.65
```



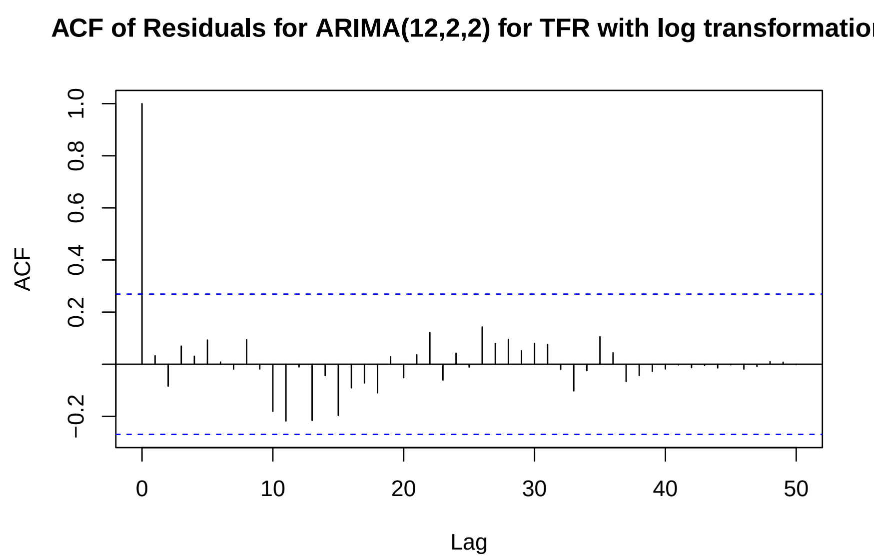


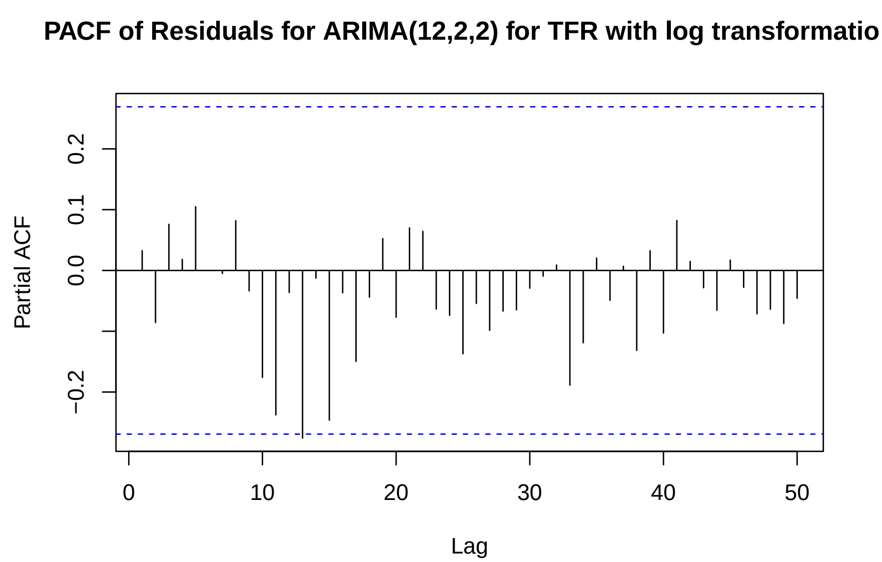
```
## Series: log(TFR_ts)
## ARIMA(13,2,2)
##
## Coefficients:
##           ar1      ar2      ar3      ar4     ar5     ar6     ar7     ar8
##        0.6514  -0.2412  -0.0063  -0.1608  0.0605  0.0715  0.0280  0.1159
## s.e.   0.1194   0.1202   0.1193   0.1180  0.1158  0.1095  0.1099  0.1125
##            ar9     ar10     ar11    ar12     ar13      ma1     ma2
##        -0.1105  -0.0649  -0.1966  0.4240  -0.6274  -1.6346  0.7851
## s.e.    0.1156   0.1159   0.1124  0.1126   0.0987   0.1446  0.1392
##
## sigma^2 = 0.002717:  log likelihood = 82.19
## AIC=-132.39   AICc=-116.39   BIC=-101.48
```





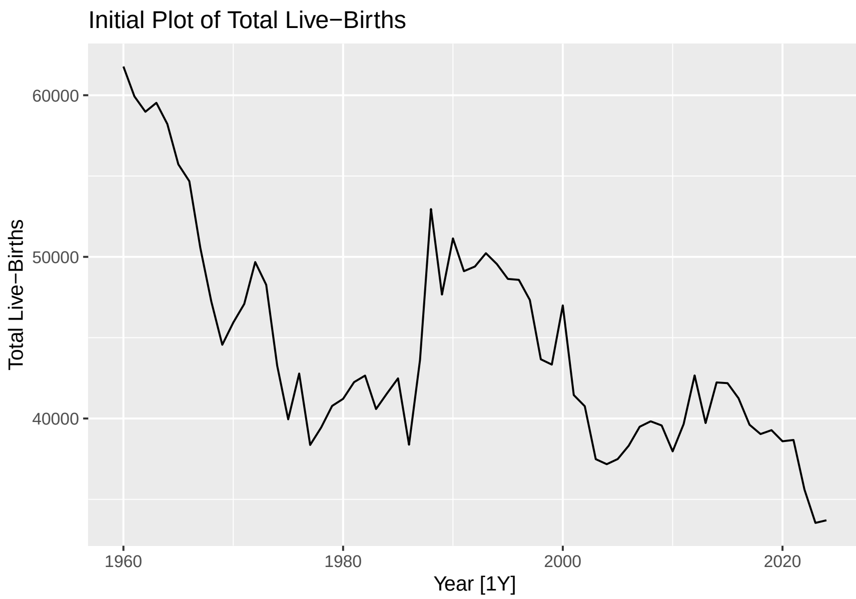
```
## Series: log(TFR_ts)
## ARIMA(12,2,2)
##
## Coefficients:
##           ar1      ar2      ar3      ar4      ar5      ar6      ar7     ar8
##       -0.2761  -0.7583  -0.6170  -0.6697  -0.4488  -0.2522  -0.1080  0.1602
## s.e.   0.1485   0.1781   0.1949   0.2095   0.2239   0.2303   0.2283  0.2136
##          ar9    ar10    ar11    ar12      ma1     ma2
##       0.1756  0.2301  0.1476  0.5613  -0.6465  0.6230
## s.e.  0.1867  0.1625  0.1233  0.1146   0.1780  0.2623
##
## sigma^2 = 0.003176:  log likelihood = 78.67
## AIC=-127.34   AICc=-113.62   BIC=-98.36
```

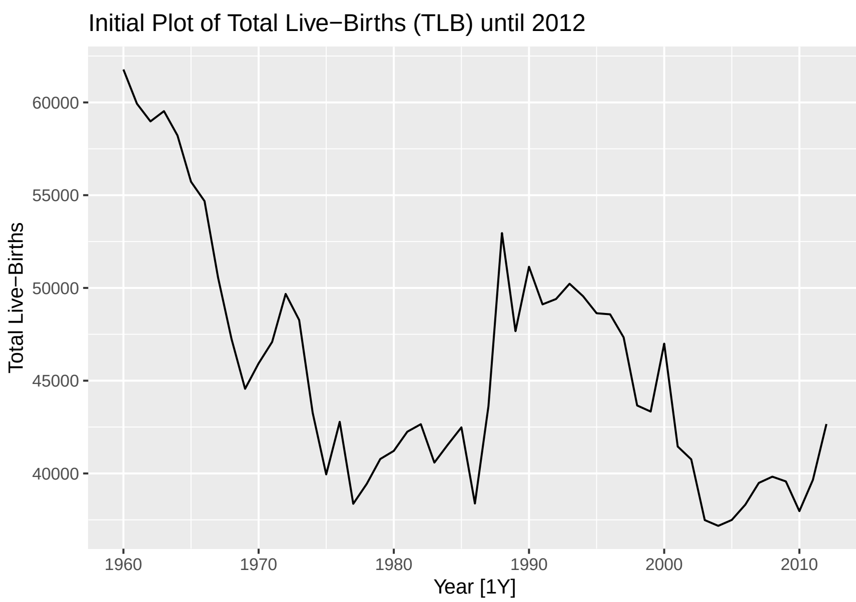




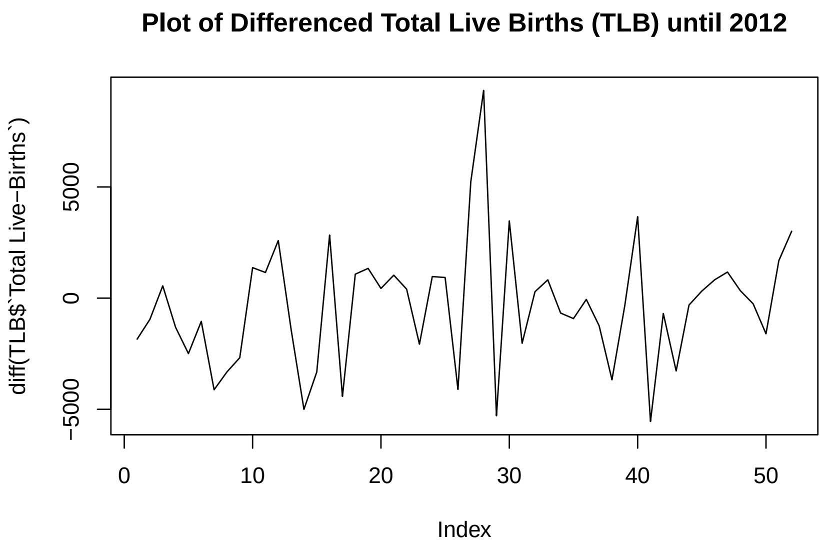
The ACF for TFR shows significant positive autocorrelations from lags 1 to 9, while TLB shows significant positive autocorrelations from lags 1 to 6, with a slow decrease in the ACF as the lag increases. This suggests that neither series is white noise, as there are multiple spikes outside the significance bounds.

The presence of a strong trend and slow decrease confirms that both TFR and TLB are non-stationary series. The absence of a sharp cut-off suggests that differencing or an autoregressive (AR) model may be suitable for this series. An extended ACF plot for TFR confirms that significant positive autocorrelations persist up to lag 10.

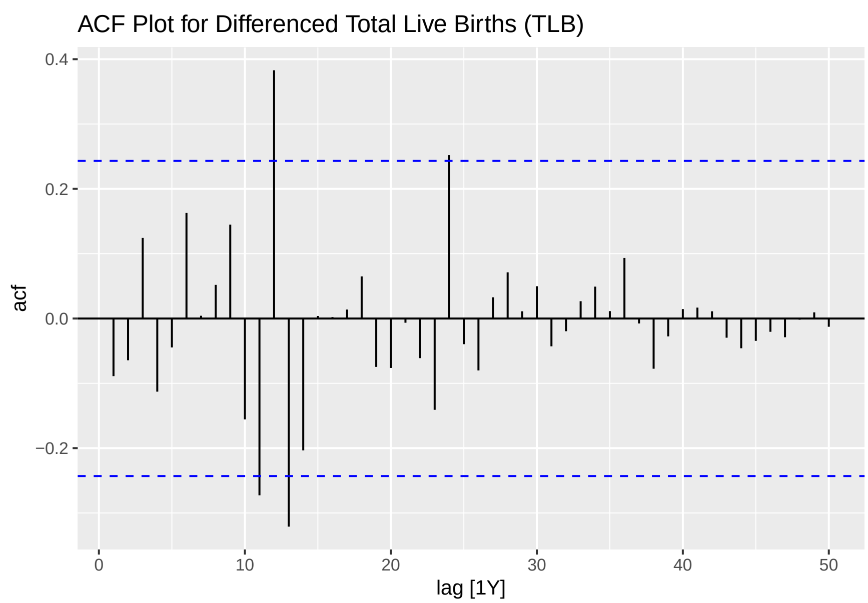


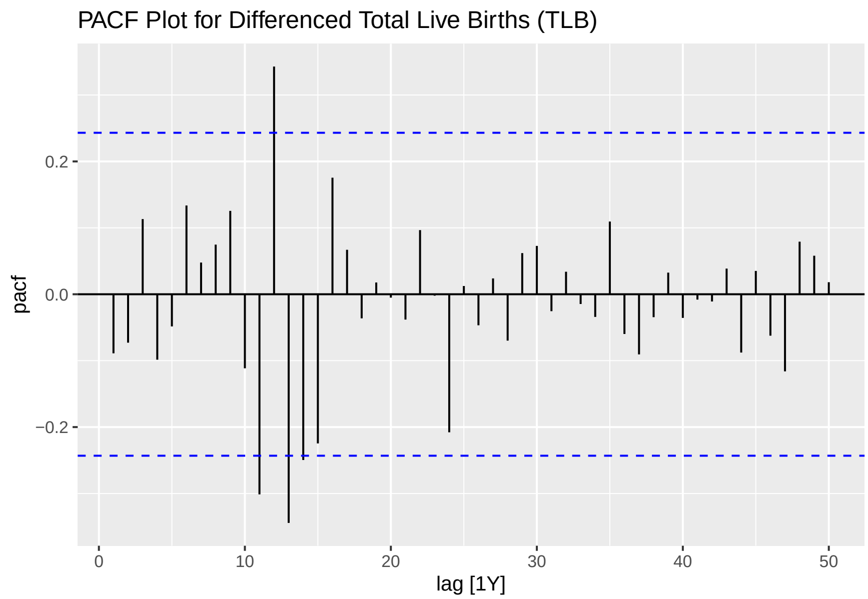


From the initial plots of TFR and TLB, both series show a long-term decreasing trend over time, with TFR exhibiting a more pronounced decline. A notable change is observed around 1980 to 1990, where both series display a temporary broad peak before continuing their downward trend. As the data is yearly, seasonality cannot be determined. While both series may suggest cyclical behaviour, the trend would need to be removed to confirm this.

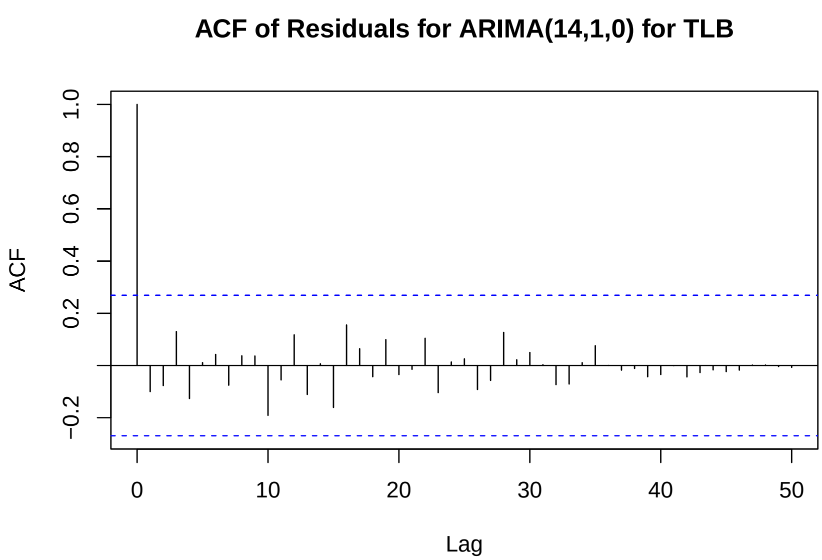


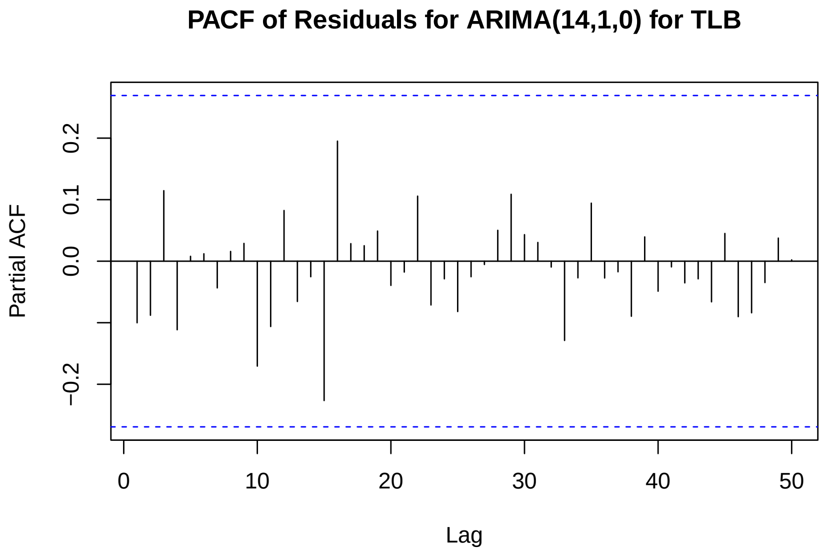

```
##   kpss_stat kpss_pvalue
##   0.2126452   0.1000000
```



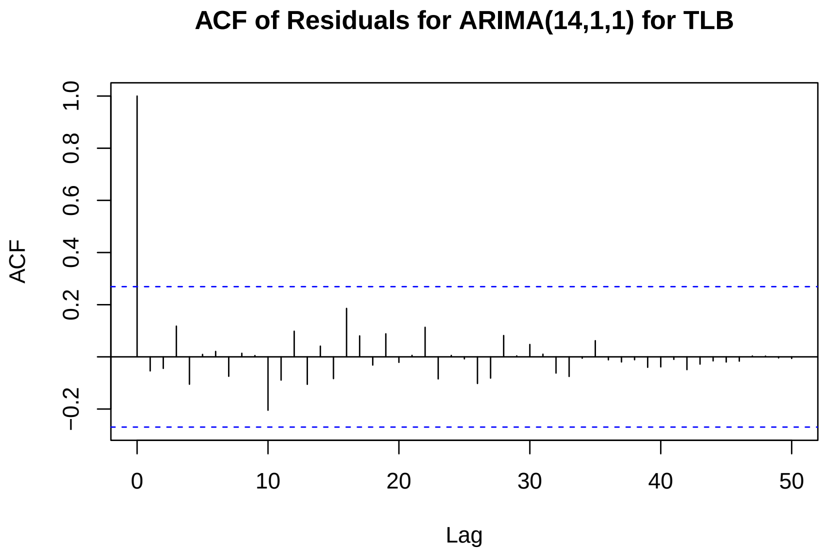


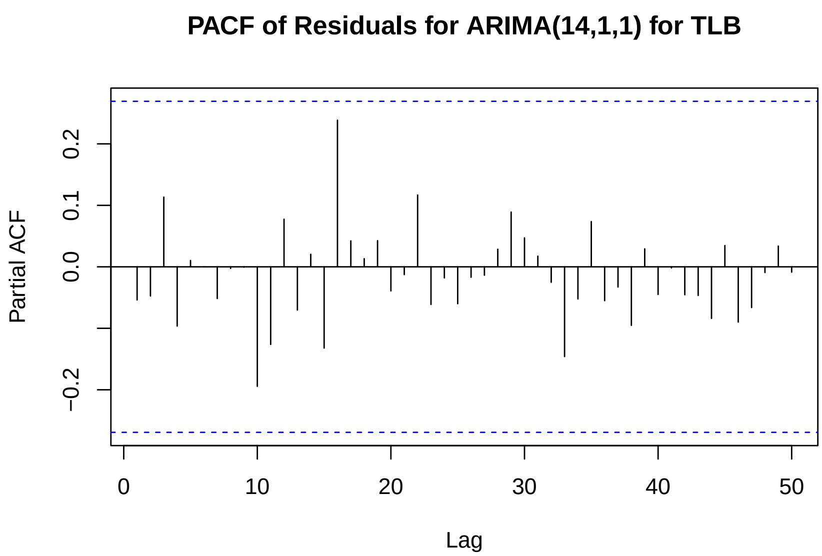

```
## Series: TLB_ts
## ARIMA(14,1,0)
##
## Coefficients:
##           ar1     ar2      ar3      ar4     ar5     ar6     ar7     ar8     ar9
##        0.0260  0.0380  -0.0184  -0.0555  0.0794  0.1459  0.1111  0.1666  0.0608
## s.e.   0.1344  0.1218   0.1091   0.1027  0.1050  0.1015  0.1087  0.1051  0.1087
##           ar10     ar11    ar12     ar13     ar14
##        -0.0696  -0.2211  0.2823  -0.3715  -0.2759
## s.e.    0.1085   0.1078  0.1195   0.1245   0.1466
##
## sigma^2 = 5629775:  log likelihood = -472.66
## AIC=975.33   AICc=988.66   BIC=1004.6
```



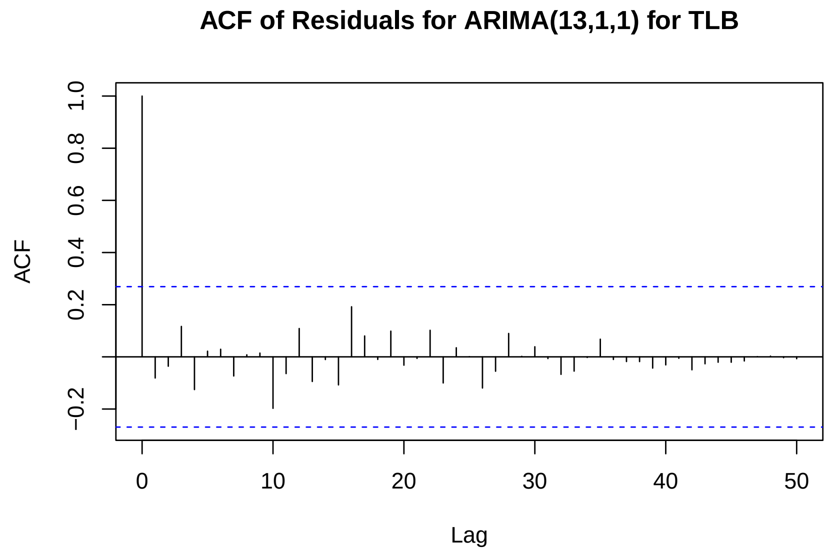

PACF of Residuals for ARIMA(14,1,0) for TLB
Partial ACF
−0.2
0.0
0.1
0.2
0
10
20
30
40
50
Lag

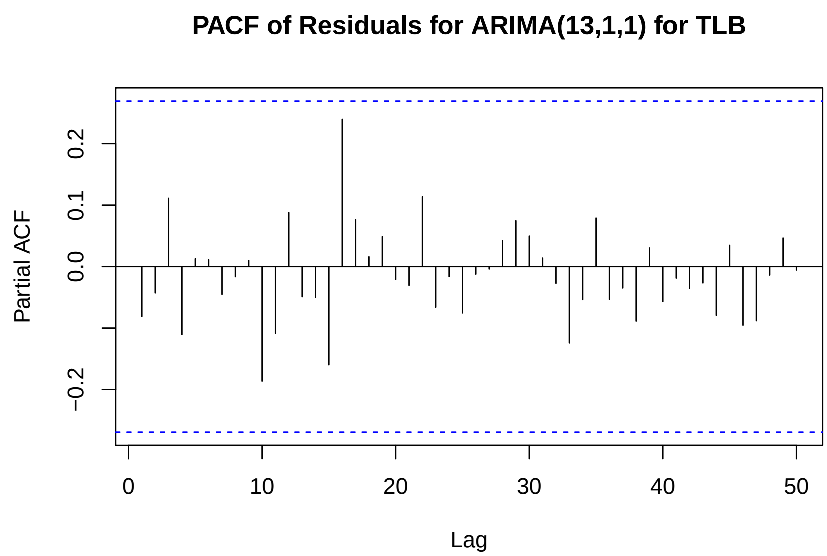
```
## Series: TLB_ts
## ARIMA(14,1,1)
##
## Coefficients:
##          ar1     ar2      ar3      ar4     ar5     ar6     ar7     ar8     ar9
##       0.3221  0.0119  -0.0149  -0.0796  0.0884  0.1294  0.0755  0.1342  0.0115
## s.e.  0.2642  0.1325   0.1107   0.1046  0.1040  0.1011  0.1084  0.1061  0.1111
##          ar10     ar11    ar12     ar13     ar14      ma1
##       -0.0862  -0.1932  0.3567  -0.4616  -0.2020  -0.3636
## s.e.   0.1062   0.1048  0.1256   0.1418   0.1864   0.2464
##
## sigma^2 = 5507699:  log likelihood = -471.81
## AIC=975.62   AICc=991.17   BIC=1006.84
```



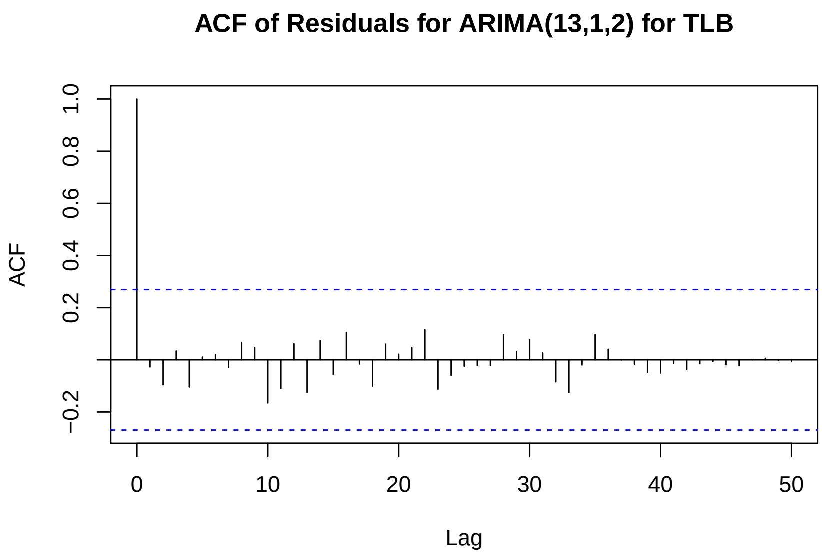


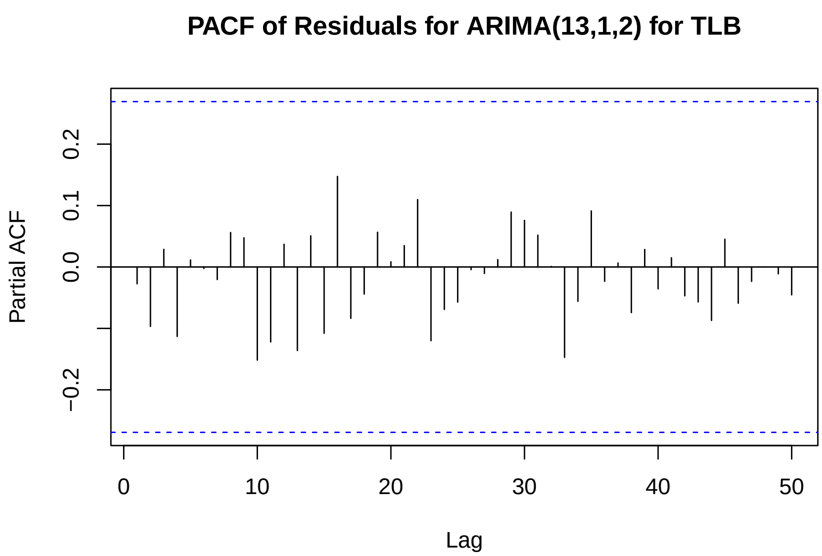
```
## Series: TLB_ts
## ARIMA(13,1,1)
##
## Coefficients:
##          ar1      ar2     ar3      ar4     ar5     ar6     ar7     ar8      ar9
##       0.5571  -0.0703  0.0264  -0.0762  0.0940  0.1004  0.0542  0.1077  -0.0216
## s.e.  0.1677   0.1156  0.1116   0.1120  0.1104  0.1040  0.1090  0.1095   0.1125
##          ar10     ar11    ar12     ar13      ma1
##       -0.0853  -0.1964  0.4091  -0.5605  -0.5344
## s.e.   0.1130   0.1112  0.1223   0.1126   0.1886
##
## sigma^2 = 5503186:  log likelihood = -472.28
## AIC=974.57   AICc=987.9   BIC=1003.83
```





```
## Series: TLB_ts
## ARIMA(13,1,2)
##
## Coefficients:
##          ar1      ar2      ar3      ar4     ar5     ar6     ar7     ar8
##       0.6648  -0.4027  -0.0077  -0.0835  0.1408  0.0861  0.0430  0.1468
## s.e.  0.1376   0.1968   0.1260   0.1252  0.1252  0.1203  0.1199  0.1246
##          ar9     ar10     ar11    ar12     ar13      ma1     ma2
##      -0.0053  -0.0478  -0.1305  0.4076  -0.6385  -0.7326  0.5855
## s.e.  0.1262   0.1271   0.1253  0.1159   0.1041   0.1611  0.3171
##
## sigma^2 = 5122169:  log likelihood = -470.51
## AIC=973.01   AICc=988.56   BIC=1004.23
```





Results

TFR Model	AIC	AICc	MSE	MAE
ARIMA(16,1,0)	-63.99	-45.99	0.04753075	0.1690442
ARIMA(16,1,1)	-62.39	-42.66	0.04783471	0.1700046
ARIMA(15,1,1)	-60.26	-42.26	0.08431368	0.223524
ARIMA(14,1,3)	-60.99	-40.26	0.01899351	0.09964934
ARIMA(12,2,3)	-62.57	-46.57	0.01218546	0.08274547
ARIMA(11,2,3)	-52.91	-39.19	0.05260825	0.1823268
ARIMA(10,2,3)	-54.69	-43.02	0.05680021	0.1898403

Log Transformation

TFR Model	AIC	AICc	MSE	MAE
ARIMA(13,1,3)	-136	-118	0.02602064	0.121592
ARIMA(12,1,4)	-131.83	-113.83	0.02101819	0.1132223
ARIMA(13,1,5)	-133.96	-110.21	0.05826379	0.1840232
ARIMA(12,1,5)	-130.02	-109.3	0.02172126	0.1147922
ARIMA(15,2,0)	-133.67	-117.67	0.01869052	0.1056717
ARIMA(15,2,2)	-135.54	-114.17	0.02235162	0.1187815
ARIMA(14,2,2)	-130.49	-111.94	0.01087075	0.08352761
ARIMA(13,2,2)	-132.39	-116.39	0.007426324	0.06826933
ARIMA(12,2,2)	-127.34	-113.62	0.0498791	0.1720094

TLB Model	AIC	AICc	MSE	MAE
ARIMA(14,1,0)	975.33	988.66	67367429	6853.008
ARIMA(14,1,1)	975.62	991.17	84179161	7769.997
ARIMA(13,1,1)	974.57	987.9	82856991	7698.072
ARIMA(13,1,2)	973.01	988.56	54382525	6014.802

Discussion

Conclusion

References

Appendices / Statisical Appendix

Research Questions

Can the ARIMA model accurately forecast the Singapore's Total Live Births (TLB) and Total Fertility Rate (TFR) from 2013 to 2024 given the data from 1960 to 2012, capturing the short-term fluctuations?

Statistical Appendix

The following statistical methods were applied throughout the analysis in the order presented below.

Autoregressive Model (AR)

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \varepsilon_t$$

where:

y_t is the value at time t

ϕ_i are model parameters

ε_t is white noise

Models the current values using its previous values (lags)

Residuals

$$e_t = y_t - \hat{y}_t$$

y_t is the actual observed value at time t

$\hat{y}_t$ is the predicted value at time t

The error which is the difference between the actual observed value and predicted value at time t, used to check the chosen model accounts for all the patterns

Autocorrelation Function (ACF)

$$\rho_k = \frac{\mathrm{Cov}(y_t, y_{t-k})}{\mathrm{Var}(y_t)}$$

where:

ρ_k is the autocorrelation at lag k

Measures the correlation between a series y_t and lagged values y_{t-k}, which checks for trend, seasonality and whether residuals are white noise

Box-Cox Transformation

$$y^{(\lambda)} = \begin{cases} \frac{y^\lambda - 1}{\lambda}, & \lambda \neq 0 \\ \log(y), & \lambda = 0 \end{cases}$$

λ was estimated using the Guerrero method

STL Decomposition

$$y_t = T_t + R_t$$

where:

T_t is the trend component,

R_t is the remainder

Decomposes time series into trend and remainder components, usually with the seasonality component as well but the data in this analysis is yearly

Moving Average (MA)

$$MA_k = \frac{1}{k} \sum_{i=-(k-1)/2}^{(k-1)/2} y_{t+i}$$

k is the window size

In this analysis when k = 5, the moving average at time t is calculated using the 2 observations before and 2 observations after t. This process smooths out short-term fluctuations in the series

https://github.com/twalepear/TSA.git